
PRe Report

*AI & numerical computations of
path functionals on Wiener space*

Maitre de stage : Toshihiro Yamada

Élève : Leto Van Ruymbeke

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Abstract

Pricing a financial product amounts to computing an expectation of the form

$$\mathbb{E}^{\mathbb{P}} \left[f \left((X_t)_{t \in [0, T]} \right) \right]$$

where $(X_t)_t$ is the solution of a stochastic differential equation and f is a payoff that may read the whole trajectory and not only its final value. Outside of a handful of textbook cases, this expectation has no closed form, and the standard answer is a Monte-Carlo simulation run on a **discretised** version of X – an Euler scheme, most of the time. This is exactly where the difficulty lies : discretising introduces a **bias** that no number of samples will ever remove. Only a finer time grid can, at a computational cost, and never entirely. Worse, that bias is invisible from inside the simulation : nothing in the output tells us how far the answer sits from the price we are after. Sometimes, decreasing relentlessly the step size can even harm the accuracy of the final result, due to the shrinking of the confidence interval of the validating test. A very welcome surprise of this internship is that stochastic calculus is one of the rare mathematical area where **exact simulation** can replace discretisation.

Thus, the methods studied during this internship take the opposite route. Instead of making the grid finer, they **remove the time grid altogether** and replace it by the exact simulation of a few well-chosen random variables [1, 5]. What is left is an estimator carrying no bias at all, and only a variance – and a variance is something that samples do reduce, at a rate we know. The price to pay is mathematical : each of these methods rests on a change of variable, a change of probability, and a fair amount of care about the hypotheses under which they hold. Path dependency is what makes the exercise delicate : a barrier option does not only look at X_T , but at $\min_{t \leq T} X_t$, a functional of the entire path that a discrete grid is structurally unable to observe between two of its points.

This work was carried out under the supervision of Professor Toshihiro Yamada, on the numerical computation of path functionals on the Wiener space. The question is not confined to finance : simulating a diffusion without discretisation bias is a concern shared by statistics, physics and biology, and it is also, in the

longer run, an artificial intelligence question. The machine-learning solvers now used for high-dimensional partial differential equations are trained on simulated trajectories ; a biased simulator quietly biases the very target the network is asked to learn, while the variance of the estimator decides how much data the training needs. Unbiased, grid-free estimators are therefore a natural building block for those approaches, and this report deals with the probabilistic and numerical foundations they rest on [19, 21].

This report gathers the three steps of the project. **Chapter I** builds an unbiased Monte-Carlo method – *Beskos-Roberts* – for $\mathbb{E}^{\mathbb{P}} \left[f \left((X_t)_{t \in [0, T]} \right) \right]$, first on a European call under Black-Scholes, then on a **barrier** call. **Chapter II** keeps that barrier payoff but changes the model for (CEV), and derives an equation meant to reduce the variance of its numerical computation. **Chapter III** comes back to the general setting and gives a new formula for the barrier price which, contrary to Chapter I, does not ask the function c to be bounded.

Framework

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and $(W_t)_t$ is supposed to be a standard Brownian motion with respect to the filtration $(\mathcal{F}_t)_t$.

Chapter I

We consider a diffusion process $(X_t)_t$ that verifies :

$$(\text{SDE}) \quad \begin{cases} dX_t = b(X_t)dt + \sigma(X_t)dW_t \\ X_0 = x \end{cases}$$

We want to compute $\mathbb{E}^\mathbb{P} \left[f \left((X_t)_{t \in [0, T]} \right) \right]$ for a given function f .

b and σ are supposed to be “appropriate” : the exact conditions they have to respect are determined in the **Sufficient Conditions part**.

I Equivalent equation, constant volatility

$(X_t)_t$ is the solution of (SDE), where we assume b and σ respect the sufficient conditions for that solution to **exist** and be **unique**. Under those, X_t is an Itô process. We define the *Lamperti transform* [6] $Y_t = g(X_t)$, where $g(x) = \int^x \frac{1}{\sigma(y)} dy$.

We also assume $g \in C^2$. By Itô formula:

$$dY_t = g'(X_t)dX_t + \frac{1}{2}g''(X_t)\sigma^2(X_t)dt$$

where $g'(x) = \frac{1}{\sigma(x)}$ and $g''(x) = -\frac{\sigma'(x)}{\sigma^2(x)}$.

It yields :

$$\begin{aligned} dY_t &= \frac{1}{\sigma(X_t)}(b(X_t)dt + \sigma(X_t)dW_t) - \frac{1}{2} \frac{\sigma'(X_t)}{\sigma^2(X_t)} \sigma^2(X_t)dt \\ &= \left(\frac{b(X_t)}{\sigma(X_t)} - \frac{1}{2} \sigma'(X_t) \right) dt + dW_t \end{aligned}$$

We deduce :

$$dY_t = \left[\frac{b(g^{-1}(Y_t))}{\sigma(g^{-1}(Y_t))} - \frac{1}{2} \sigma'(g^{-1}(Y_t)) \right] dt + dW_t$$

from which we define $a(y) = \frac{b(g^{-1}(y))}{\sigma(g^{-1}(y))} - \frac{1}{2} \sigma'(g^{-1}(y))$

Sufficient conditions

Let's now define what we mean by "appropriate" for b and σ . Rather than piling up conditions on b and σ separately, we state them, as far as possible, on the two objects the construction really uses : the transform g , and the drift a of Y :

- **Existence and uniqueness of X_t** - there exists K such that, $\forall x, y$:
 - *Lipschitz* : $|b(x) - b(y)| + |\sigma(x) - \sigma(y)| \leq K|x - y|$
 - *Linear growth* : $|b(x)| + |\sigma(x)| \leq K(1 + |x|)$
- $g \in C^2$: We need σ' to be continuous and σ^2 not vanishing. We take the sufficient condition : $\begin{cases} \sigma \in C^1 \\ \forall x, \sigma(x) \neq 0 \end{cases}$
- g^{-1} **existence** : g must be a bijection on its domain. We thus need g to be monotone $\iff g' > 0$ (or $g' < 0$) $\iff \sigma > 0$ (or $\sigma < 0$), which is always the case since we supposed that σ is continuous **and** doesn't vanish.
- $a \in C^1$: the drift a of Y , defined at the end of the previous part, has to be C^1 , so that its primitive h used in Section III is C^2 and Itô formula applies to $h(Y_t)$. Asking $b \in C^1$ and $\sigma \in C^2$ is enough.

Conclusion : from now on we keep the following as verified :

- b and σ are K -Lipschitz with linear growth of weight K , which makes X_t exist and be unique ;
- $\sigma \in C^2, \sigma > 0$ and $b \in C^1$, which make g, g^{-1} and a well defined, with $g \in C^2$ and $a \in C^1$.

They allow Y_t to exist. From that point on, every further hypothesis of this report is stated on a alone : it is the only function the following computations actually see.

II Girsanov : drift removal by change of probability

In order to apply *Girsanov-Maruyama* [12, 11], we need :

1. $\int_0^T a(Y_t)^2 dt < \infty$ almost surely. It is granted as soon as a is bounded on

$g\left(\text{Ran}_\omega(X)\right)$, where $\text{Ran}_\omega(X) = \{X_t(\omega), t \in [0, T]\}$ is the range of values X visits, which still needs to be determined. We will see that the second condition below is in fact enough to get it.

2. $L_t = \exp\left(-\int_0^t a(Y_s) dW_s - \frac{1}{2} \int_0^t a(Y_s)^2 ds\right)$ to be a $\mathbb{P}$ -martingale.

Here is the proof that $(L_t)_t$ is indeed a martingale, and under which conditions.

Let's define $Z_t = -\int_0^t a(Y_s) dW_s - \frac{1}{2} \int_0^t a(Y_s)^2 ds$. We want to compute dL_t .

As $L_t = \exp(Z_t)$, by Itô formula :

$$\begin{aligned} dL_t &= \exp(Z_t) dZ_t + \frac{1}{2} \exp(Z_t) d\langle Z, Z \rangle_t \\ &= L_t \left(-a(Y_t) dW_t - \frac{1}{2} a^2(Y_t) dt \right) + \frac{1}{2} L_t a^2(Y_t) dt \\ &= -a(Y_t) L_t dW_t \end{aligned}$$

$$\text{So } L_t = L_0 + \int_0^t (-a(Y_s) L_s) dW_s = 1 + \int_0^t (-a(Y_s) L_s) dW_s$$

So L_t is a local martingale, and with the next additional condition that we consider to be verified, L_t is a $\mathbb{P}$ -martingale :

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T a^2(Y_s) ds \right) \right] < +\infty$$

That last condition also ensures $\int_0^T a(Y_t)^2 dt < \infty$ almost surely, since a random variable with a finite expectation is almost surely finite. We therefore keep it as the single hypothesis under which both *Girsanov-Maruyama* requirements hold. According to this theorem, under $\mathbb{Q}$ of density L_T with respect to $\mathbb{P}$:

$$W_t^{\mathbb{Q}} = W_t + \int_0^t a(Y_s) ds \text{ is a standard Brownian motion}$$

But as we know, $\mathbb{E}^{\mathbb{P}}[f((X))] = \mathbb{E}^{\mathbb{P}}[f(g^{-1}(Y))]$ and $\frac{d\mathbb{Q}}{d\mathbb{P}} = L_T$, which means :

$$\begin{aligned}\forall A \in \mathcal{F}, \quad \mathbb{Q}(A) &= \int_A L_T d\mathbb{P}(\omega) \text{ by definition of } L_T \\ &= \mathbb{E}^{\mathbb{P}}[L_T \mathbf{1}_A]\end{aligned}$$

Thus, for all random variable $Z : \Omega \rightarrow \mathbb{R}$,

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[Z] &= \int_{\Omega} Z(\omega) d\mathbb{Q}(\omega) \\ &= \int_{\Omega} Z(\omega) L_T d\mathbb{P}(\omega) \\ &= \mathbb{E}^{\mathbb{P}}[L_T Z]\end{aligned}$$

and if $L_T > 0$, which is the case here : $\mathbb{E}^{\mathbb{P}}[Z] = \mathbb{E}^{\mathbb{Q}}\left[\frac{Z}{L_T}\right]$.

Now we take $Z := f(g^{-1}(Y))$:

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}[f(X)] &= \mathbb{E}^{\mathbb{Q}}\left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s + \frac{1}{2} \int_0^T a^2(Y_s) ds}\right] \\ &= \mathbb{E}^{\mathbb{Q}}\left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s^{\mathbb{Q}} - \frac{1}{2} \int_0^T a^2(Y_s) ds}\right]\end{aligned}$$

$$\text{since } dW_t^{\mathbb{Q}} = dW_t + a(Y_t)dt$$

Note : the paper given by Yamada-sensei seems to have used dW_t when $dW_t^{\mathbb{Q}}$ is intended for this last result. But the measure change $dW_t \rightarrow dW_t^{\mathbb{Q}}$ requires a $-\frac{1}{2}$ instead of a $\frac{1}{2}$. The final equation being :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}}\left[f(g^{-1}(Y)) \cdot e^{\int_0^T a(Y_s) dW_s^{\mathbb{Q}} - \frac{1}{2} \int_0^T a^2(Y_s) ds}\right]$$

Sufficient conditions update

We add to the Sufficient Conditions found before :

- **Girsanov-Maruyama condition** : $\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T a^2(Y_s) ds \right) \right] < +\infty$

Note : we state this condition on a , and not on b and σ . One can of course give conditions on b and σ that imply it – for instance b and σ' bounded together with $\sigma \geq \alpha > 0$ – but they are needlessly strong : neither Black-Scholes ($b(x) = rx$ is unbounded and $\sigma(x) = \sigma_0 x$ vanishes at 0), nor the **(CEV)** model of Chapter II, satisfies them, while both have a bounded a – constant, even, for Black-Scholes, where $a \equiv \frac{r}{\sigma_0} - \frac{\sigma_0}{2}$ – so that the condition above holds trivially. Stating the hypotheses on a keeps them at the level where they are actually used.

III Simplifying implementation : $dW^{\mathbb{Q}}$ integral removal

The presence of the integral with respect to $dW_t^{\mathbb{Q}}$ is a hindrance to computation. In practice, deterministic integrals are easy to approximate but stochastic ones need us to discretize Brownian motions and approximate Itô integrals. Thus we call h a primitive of a , which is C^2 since $a \in C^1$, as required for Itô formula to apply to $h(Y_t)$ below.

We define $Z_t = h(Y_t)$. By Itô formula :

$$dZ_t = h'(Y_t)dY_t + \frac{1}{2}h''(Y_t)d\langle Y, Y \rangle_t$$

$$\begin{aligned} \text{while } dY_t &= a(Y_t)dt + dW_t \\ &= a(Y_t)dt + dW_t^{\mathbb{Q}} - a(Y_t)dt \\ &= dW_t^{\mathbb{Q}} \end{aligned}$$

Thus, $(Y_t)_t$ is a Brownian motion under $\mathbb{Q}$, and $d\langle Y, Y \rangle_t = dt$, so :

$$\begin{aligned} dZ_t &= a(Y_t)dW_t^{\mathbb{Q}} + \frac{1}{2}a'(Y_t)dt \\ \Rightarrow Z_T - Z_0 &= \int_0^T a(Y_s)dW_s^{\mathbb{Q}} + \frac{1}{2} \int_0^T a'(Y_s)ds \\ \Rightarrow h(Y_T) - h(Y_0) &= \int_0^T a(Y_s)dW_s^{\mathbb{Q}} + \frac{1}{2} \int_0^T a'(Y_s)ds \\ \Rightarrow \int_0^T a(Y_s)dW_s^{\mathbb{Q}} &= h(Y_T) - h(Y_0) - \frac{1}{2} \int_0^T a'(Y_s)ds \end{aligned}$$

In the end, we have :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T -\frac{1}{2}(a'(Y_s) + a(Y_s)^2)ds} \right]$$

from which we define $c := -\frac{1}{2}(a' + a^2)$:

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s)ds} \right]$$

IV Code

1 Monte-Carlo under Euler scheme

a) Theory

First we implement the function giving the exact Black-Scholes price for a European call, in *black_scholes.py*.

In *models.py*, we define a *Python* Class named *BSmodel* which represents the classic Black-Scholes SDE with :

- S_0 , the starting point
- $b : x \mapsto rx$ the *drift*
- $\sigma : x \mapsto \sigma x$ the *volatility*

Then, in *monte_carlo.py*, we define a first Monte-Carlo function that, given a model, a maturity, a payoff function, a number of steps and a number of paths, discretizes the model using Euler scheme [7, 9] to estimate $E[f(X)]$ and $\text{Var}(f(X))$ with the usual estimators.

The induction gives :

$$\begin{aligned}\hat{X}_{k+1} &= \hat{X}_k + b(\hat{X}_k)\Delta t + \sigma(\hat{X}_k)\sqrt{\Delta t}Z_k \quad (*) \\ &= \hat{X}_k(1 + r\Delta t + \sigma\sqrt{\Delta t}Z_k) \text{ for B-S solution}\end{aligned}$$

where $\Delta t = \frac{T}{N}$ and $Z_k \sim \mathcal{N}(0, 1)$

A few things to note:

- This method gives us $\hat{\mu}_M = \frac{1}{M} \sum_{j=1}^M f(\hat{X}_j)$
- The variance of this method is $\hat{\sigma}_M^2 = \text{Var}(\hat{\mu}_M) = \sigma_{\text{Euler}}^2/M$, where $\sigma_{\text{Euler}}^2 = \text{Var}(f(\hat{X}))$.
- We have the exact induction equation :

$$X_{k+1} = X_k \cdot \exp\left(\left(r - \sigma^2 \frac{1}{2}\right)\Delta t + \sigma\sqrt{\Delta t} \cdot Z_k\right)$$

because the solution of B-S is $X_t = S_0 e^{(r - \frac{\sigma^2}{2})t + \sigma W_t}$. Thus, (*) is just a Taylor expansion at $O(\Delta t)$ ($= O(\frac{1}{N})$).

- The exact solution and the approximation differ, **in conditional expectation**, from Taylor expansion at order Δt^2 and higher. r increases that difference. Indeed :

$$\begin{aligned}\mathbb{E}[X_{k+1}|X_k] &= X_k \cdot \mathbb{E}\left[\exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}Z_k\right)\right] \\ &= X_k \exp\left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t\right) \cdot \mathbb{E}\left[e^{\sigma\sqrt{\Delta t}Z_k}\right] \\ &= X_k e^{(r - \frac{1}{2}\sigma^2)\Delta t} e^{\frac{1}{2}\sigma^2\Delta t} \\ &= X_k e^{r\Delta t}\end{aligned}$$

$$\mathbb{E}[\hat{X}_{k+1}|\hat{X}_k] = \hat{X}_k(1 + r\Delta t)$$

$$\Rightarrow \mathbb{E}[X_{k+1}|X_k] - \mathbb{E}[\hat{X}_{k+1}|\hat{X}_k] = \frac{r^2}{2}X_k\Delta t^2 + O(\Delta t^3)$$

- Finally, as for any $p > 0$, $O(\Delta t^p) = O(\frac{1}{N^p})$, there exists a **bias** between the exact and the Euler solution that decreases when the grid becomes denser. A local error of order Δt^2 repeated over the $N = T/\Delta t$ steps of the grid

aggregates into a global bias of order $\Delta t = O(1/N)$, which is the weak order 1 we measure numerically in Figure 2.

b) First tests' results

In *main.py*, we started by implementing a function named *validate*. It takes a set of parameters m, f, T, N, M and *ref* (see the code for more details), computes both $\hat{\mu}_M$ and $\hat{\sigma}_M$ defined earlier. Finally, *validate* returns true if the validating condition is verified :

$$|\mu - \hat{\mu}_M| \leq 4\hat{\sigma}_M$$

We define n_σ by :

$$n_\sigma = \frac{|\mu - \hat{\mu}_M|}{\hat{\sigma}_M}$$

We can show that there is a 0.0063% chance for a given instance not to pass this test. The reason why we need it to be so low is that the Monte-Carlo method will run hundreds of these tests for different values of the parameters, so the probability of a failure will increase. Thus **we take $4\hat{\sigma}$ as a threshold** and not $3\hat{\sigma}$ or $2\hat{\sigma}$, that would be satisfied with high probability for one test, but not for hundreds of them.

Then we run this test for a decent number of instances. As such, we executed *testMC()*, that runs *validate(m,f,T,N,M,ref)* for every possible combination of the parameters given by the next table :

Parameter	Tested values
T	1
S_0	100
r	0.01, 0.05, 0.1, 0.4
σ	0.1, 0.2, 0.3, 0.5
K	$S_0, S_0 - 20, S_0 + 20$
N	10, 100, 1000
M	$10, 10^2, 10^3, 10^4, 10^5$

Table 1: **Initial test – Monte-Carlo coarse parameters for Euler scheme**

testMC returns nothing, but prints every failure and for what specific set of parameters it occurred. This allows us to study if these failures were statistically expected, or if there is a bug in our code.

Before presenting the results, one important thing to mention is that the test held here can produce spurious failures, that is rejections with no bug behind them. Indeed, what we want is for this test to warn us when our estimation, $\hat{\mu}_M$, is too far from μ that we try to estimate. However, for big values of M (large amount of samples $X_T^{(i)}$), $\hat{\sigma}_M = \hat{\sigma}/\sqrt{M}$ decreases so the test window becomes more and more narrow. On the other hand, increasing M makes $\hat{\mu}_M$ closer to $\mu_{\text{Euler}} = \mathbb{E}[f(\hat{X})] \neq \mu$. This means that, as long as the bias $|\mu_{\text{Euler}} - \mu|$ is low, meaning the grid is fine (big N), the strictness of this test increasing with M bears no issue. If the test has too many failures, there is a high probability that there is an error in our code.

But, if the grid is too coarse (low N), the bias $|\mu_{\text{Euler}} - \mu|$ increases. Then, increasing M makes our estimator $\hat{\mu}_M$ closer to μ_{Euler} (not of μ), but also makes our test of $|\hat{\mu}_M - \mu|$ more severe. In this specific case, we can't tell whether the test rejected an instance because the code is wrong, or because M is too big compared to N .

This has to be considered when reading the test failures.

	Parameters					True Price	M-C results		Test results	
Error type	r	volatility	K	N	M	μ	$\hat{\mu}_M$	$\hat{\sigma}_M$	$ \hat{\mu}_M - \mu $ (\$-unit)	$ \hat{\mu}_M - \mu $ ($\hat{\sigma}_M$ -unit)
Type 1	0.4	0.1	100	10	10^4	49.1826	47.9543	0.1434	1.23	8.57σ
	0.4	0.1	100	10	10^5	49.1826	48.0467	0.0452	1.14	25.13σ
	0.4	0.1	80	10	10^4	69.1825	68.1714	0.1417	1.01	7.14σ
	0.4	0.1	80	10	10^5	69.1825	68.0241	0.0452	1.16	25.63σ
	0.4	0.1	120	10	10^4	29.2522	28.1350	0.1401	1.12	7.97σ
	0.4	0.1	120	10	10^5	29.2522	28.0998	0.0445	1.15	25.90σ
	0.4	0.2	100	10	10^5	49.3891	48.0755	0.0895	1.31	14.68σ
	0.4	0.2	80	10	10^4	69.1880	67.9651	0.2850	1.22	4.29σ
	0.4	0.2	80	10	10^5	69.1880	68.1753	0.0907	1.01	11.17σ
	0.4	0.2	120	10	10^5	31.0551	29.8818	0.0826	1.17	14.20σ
	0.4	0.3	100	10	10^5	50.7238	49.6940	0.1308	1.03	7.87σ
	0.4	0.3	80	10	10^4	69.4062	67.3628	0.4276	2.04	4.78σ
	0.4	0.3	80	10	10^5	69.4062	68.3606	0.1365	1.05	7.66σ
	0.4	0.3	120	10	10	34.6379	10.6880	5.1087	23.95	4.69σ
	0.4	0.3	120	10	10^4	34.6379	33.1092	0.3719	1.53	4.11σ
	0.4	0.3	120	10	10^5	34.6379	33.4654	0.1186	1.17	9.89σ
	0.4	0.5	100	10	10^5	56.3896	54.8896	0.2128	1.50	7.05σ
	0.4	0.5	100	10^3	10	56.3896	22.6136	5.5556	33.78	6.08σ
	0.4	0.5	120	10	10^4	43.5771	40.8770	0.6067	2.70	4.45σ
	0.4	0.5	120	10	10^5	43.5771	42.7537	0.1993	0.82	4.13σ
Type 2	0.01	0.1	120	10	10	0.1900	0	0	0.19	∞
	0.01	0.1	120	100	10	0.1900	0	0	0.19	∞
	0.05	0.2	120	10	10	3.4140	0	0	3.41	∞
	0.1	0.1	120	100	10	1.3276	0	0	1.33	∞
	0.1	0.2	120	100	10	5.2034	0	0	5.20	∞
Type 3	0.01	0.1	120	100	100	0.1900	0.0243	0.0242	0.17	6.85σ
	0.01	0.3	100	100	10	12.4926	1.3549	0.9215	11.14	12.09σ
	0.01	0.5	100	10^3	10	20.3469	2.6791	2.4375	17.67	7.25σ
	0.01	0.5	80	10^3	10	30.2268	4.1345	2.6078	26.09	10.01σ
	0.01	0.5	120	100	100	13.5696	5.8586	1.8098	7.71	4.26σ
	0.05	0.3	100	10^3	10	14.9609	0.3481	0.3302	14.61	44.25σ
	0.05	0.3	120	100	10	7.2580	0.5988	0.5366	6.66	12.41σ
	0.05	0.3	120	10^3	10	7.2580	1.7421	1.2793	5.52	4.31σ
	0.1	0.2	120	10^3	10	5.2034	0.9644	0.8515	4.24	4.98σ
	0.1	0.5	120	100	10	18.3132	3.8925	3.1662	14.42	4.55σ

Table 2: Monte-Carlo test failures
Failures obtained out of the 720 combinations of parameters.

c) Discussion

For 720 tests, we got ~30 failures, while we expected 0 or 1. Let's analyse these errors.

- **Type 1** : $r = 0.4$

The most common failures happen when the value of r is much above the usual values in finance : r is the riskless asset rate, it is never worth this much. But as explained in the previous remark, there is also a mathematical reason why many failures occurred for such a high value of r : the Euler scheme has a bias that increases with r . Thus, restricting to $r \leq 0.1$ for our tests is a mathematically justified decision that should avoid this type of failure, while being closer to the reality of the market.

- **Type 2** : $M = 10, K > S_0$ (OTM)

Too few samples for the Monte-Carlo bear the risk of a null standard deviation, thus leading to an infinite gap when expressed in $\hat{\sigma}_M$ -units. The test fails, when in truth a case where $\hat{\sigma}_M = 0$ shouldn't even be considered. This happens only because we have a degenerate sample : K is too high, compared to S_0 , to ensure that, among the 10 paths (too few), at least one will survive. This case teaches us that a condition has to assert that the samples are various enough. We take $M \geq 100$ to diminish the probability of $\hat{\sigma}_M = 0$.

- **Type 3** : The remaining cases

All the remaining failures share the same two features : a small M (10 or 100), and a payoff distribution that is strongly skewed to the right. Note that $N \geq 100$ in every one of those rows : contrary to **Type 1**, the Euler bias is not what makes them fail.

1. The payoff distribution is skewed to the right.

We know that $f(S_T)$ is skewed to the right because of the structure of the payoff (European call). Most of its values are around 0^+ , but a few far exceed this domain. Thus, while the median of the distribution is close to 0, $\mu = \mathbb{E}[f(S_T)]$ gets pulled to the right by the rare big values taken by the distribution. See the next graph for a better understanding.

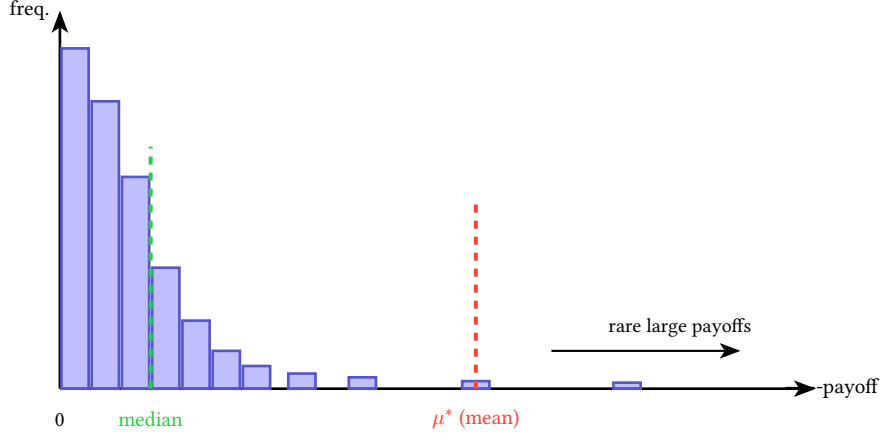


Figure 1: Payoff distribution — skewed by rare large outcomes.
The distribution is heavily concentrated near zero. A few rare but large payoffs pull the mean μ^ far to the right of the median.*

2. A small sample misses the tail, and misses it twice.

As we have a small M , we have a small sample, so a lower probability for the Euler distribution to **catch the tail** of huge values taken by the exact distribution. This leads to $\mu > \hat{\mu}_M$ in almost all cases. But the very same missing tail is also what carries most of the variance, so $\hat{\sigma}_M$, computed on that same sample, is underestimated as well. In

$$n_\sigma = \frac{|\mu - \hat{\mu}_M|}{\hat{\sigma}_M}$$

both effects push in the same direction : a numerator too big **and** a denominator too small. This is why those rows do not fail at 4σ but at 7σ , 12σ , even 44σ .

3. What matters is the effective sample size.

The quantity that governs the phenomenon is not M but the number of paths that actually contribute to the payoff, $M_{\text{eff}} := M \cdot \mathbb{P}(S_T > K) = M \cdot \Phi(d_2)$. Eight of the ten rows above have $M_{\text{eff}} \simeq 3$ to 6 : the estimator rests on a handful of non-zero samples. The two others have $M_{\text{eff}} \simeq 30$, and they are precisely the two mildest failures (4.26σ and 4.55σ), where only the skew of $\sigma = 0.5$ remains to explain the rejection.

For these reasons, the test becomes hard to pass under such conditions, independently of the correctness of the scheme. Again, increasing M

– or rather M_{eff} – is the solution. *Note* : $\hat{\mu}_M$ remains an unbiased estimator of μ_{Euler} ...

d) Second tests

The previous work taught us that :

- r must remain low, in accordance with both the reality of the market and to ensure the bias of Euler scheme doesn't prevent the results accuracy.
- M needs to be higher than 100 because a smaller M can lead to a null standard deviation which makes the tests fail in every case. Also, there is a risk that, for OTM cases, the Euler approximated variables $\hat{X}$ are too few to catch the high value taken rarely by the exact distribution, leading $\hat{\mu}_M$ to fail the test.
- N must be bigger than 100 to ensure that the Euler bias remains low.

So here are the new parameters sets :

Parameter	Tested values
T	1
S_0	100
r	0.01, 0.05, 0.1
σ	0.1, 0.2, 0.3, 0.5
K	S_0 , $S_0 - 20$, $S_0 + 20$
N	100, 1000
M	$10^2, 10^3, 10^4, 10^5$

Table 3: **Second test – Monte-Carlo adjusted parameters for Euler scheme**

By running these tests a dozen times, **no failure occurred**. The Euler scheme has been validated through suitable tests : it approximates μ in accordance with n_σ under a threshold of 4.

Moreover, we can plot the bias of the Euler scheme depending on N , under log-log scale, to confirm that our scheme bias converges in $O(\frac{1}{N})$. Here is what we get :

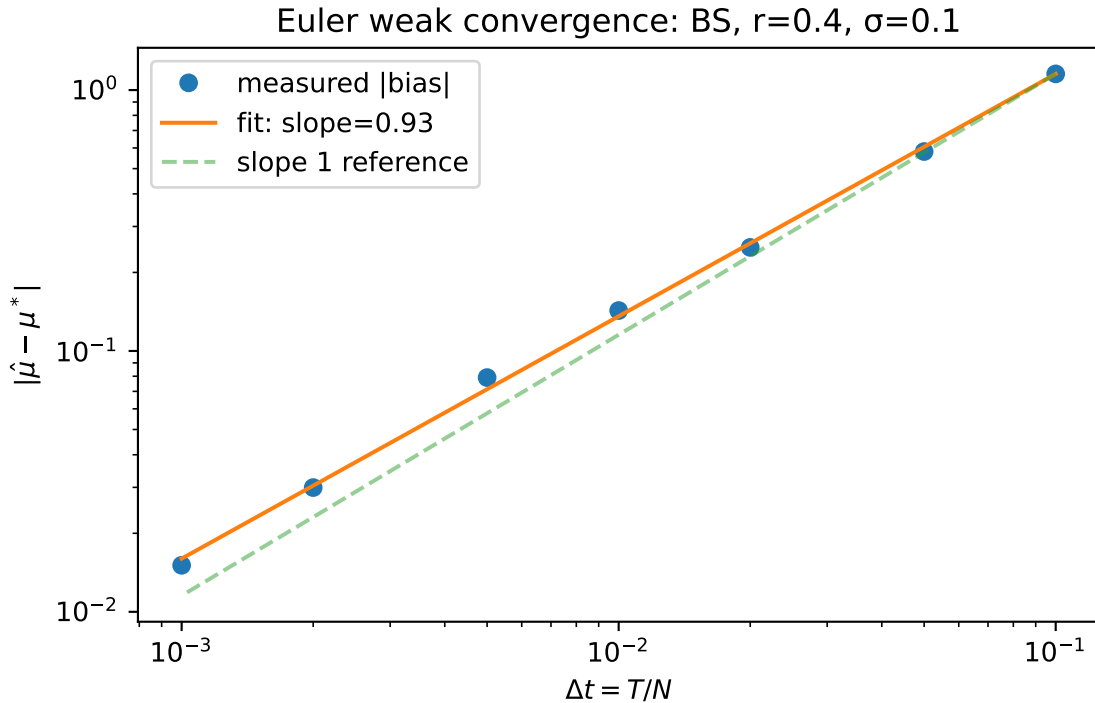


Figure 2: Euler convergence compared to $1/N$

We get a slope of 0.93 when we expected 1 : the empirical weak order of the bias is consistent with $1/N$.

2 Beskos-Roberts

a) Implementation

The previous approximation method gives us a solid way to assess the next one's accuracy and allows us to detect errors in our code. The goal now is to use the final form of $\mathbb{E}^{\mathbb{P}}[f(X)]$ we got in Section III. Let's remind it :

$$\mathbb{E}^{\mathbb{P}}[f(X)] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y)) \cdot e^{h(Y_T) - h(Y_0) + \int_0^T c(Y_s) ds} \right] (*)$$

There is one main difficulty with the implementation of this formula, that is the $e^{\int c(Y)}$: $(c(Y_s))_{s \in [0, T]}$ is a random path, which means it needs to be simulated and approximated by a scheme that would let a new bias appear. Initially, the purpose of all those steps was to find a way to compute $E[f(X)]$ that wouldn't come with any bias.

Let us define :

$$\Lambda := \int_0^T -c(Y_s) ds$$

As Λ depends on Y , it is random. But let's see it as a constant for a moment : the idea is that Λ could be the parameter of a **Poisson law**. We recall that :

$$\tilde{Z} \sim \mathcal{P}(\lambda) \iff \forall k \in \mathbb{N}, \mathbb{P}(\tilde{Z} = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Thus, if $Z \mid Y \sim \mathcal{P}(\Lambda)$, we have $\mathbb{P}(Z = 0 \mid Y) = \exp(-\Lambda) = e^{\int_0^T c(Y_s) ds}$, so we need to define $Z \mid Y$ as a Poisson law : the parameter Λ can depend on space and time, but once we fixed the path Y , Λ is fixed too.

Now, what experiment could give us a Poisson variable of parameter Λ ? We can build one using rejection sampling, which is the idea of *Beskos-Roberts* [1, 2] :

- Simulate an integer $P \sim \mathcal{P}(TB)$, where T is maturity and B is an upper bound of $-c$ (see the definition of c)
- Pick P coordinates on the time axis under $\mathcal{U}([0, T])$
- Pick P coordinates on the space axis under $\mathcal{U}([0, B])$
- That gives us a sequence of P points $(\tau_i, V_i)_{1 \leq i \leq P}$
- A point is kept when :

$$V_i < -c(Y_{\tau_i})$$

Note : this construction implicitly asks $-c$ to be **non-negative and bounded** : Λ has to be a valid Poisson parameter, and the upper bound B has to exist. We therefore add to our conditions :

$$\exists B > 0, \quad \forall y \in g\left(Ran_{\omega}(X)\right), \quad 0 \leq -c(y) \leq B$$

This is a real restriction, and it is precisely the one that **Chapter III** removes, by replacing this rejection sampling by a Poisson-weighted expansion of $e^{\int_0^T c(Y_s) ds}$.

Let us introduce this new theorem :

Theorem : Thinning

Let there be a Poisson process of intensity $\lambda(\cdot)$ from which we independently keep each point with probability $p(\cdot)$. The surviving points form a Poisson process of intensity $\lambda(\cdot)p(\cdot)$

By doing this, conditionally to a path $(Y_s)_s$, the number of surviving points $Z \mid Y$ is a Poisson random variable of parameter Λ :

- Our test has a probability of success :

$$\mathbb{P}(\{\text{no points are under } -c(Y)\} \mid Y) = \mathbb{P}(Z = 0 \mid Y)$$

- The points (τ_i, V_i) form a Poisson process of constant intensity $\lambda(t, v) = 1$ on the rectangle $[0, T] \times [0, B]$, whose total mass is TB , and we keep each of them with probability $p(t, v) = \mathbb{1}_{\{v < -c(Y_t)\}}$. By the *thinning theorem*, the kept points form a Poisson process of intensity λp , whose total mass is the area under the curve :

$$\int_0^T \int_0^B \mathbb{1}_{\{v < -c(Y_t)\}} dv dt = \int_0^T -c(Y_t) dt = \Lambda$$

Hence, $\mathbb{P}(Z = 0 \mid Y) = e^{-\Lambda}$

To get Y_{τ_i} we use a specific tool called the **Brownian bridge** [2, 3].

In our case, $Y_{\tau_i} \sim \mathcal{N}(Y_0, \tau_i)$ as proved in Chapter I part 3. However, our simulation can be much more accurate by drawing Y_T first, and then drawing the Y_{τ_i} knowing Y_T and $Y_{\tau_{i-1}}$ at each step. The idea is that, as $Y_{\tau_{i-1}}$ and Y_T are known, Y_{τ_i} follows a law of lower variance conditionally to them.

Hence the use of this method, given by the next formula. For a Brownian motion of maturity T , we have :

$$(W_t \mid W_0 = x, W_T = y) \sim \mathcal{N}\left((y - x)\frac{t}{T} + x, \frac{t(T - t)}{T}\right)$$

Eventually, our Monte-Carlo computes the wanted expectation by :

- Computing $f(g^{-1}(Y))$
- Simulating $Y_T \sim \mathcal{N}(Y_0, T)$, where $Y_0 = g(x)$: this gives $e^{h(Y_T) - h(Y_0)}$

- Simulating $P \sim \mathcal{P}(TB)$ and P independent points (τ_i, V_i) with $\begin{cases} \tau_i \sim \mathcal{U}([0, T]) \\ V_i \sim \mathcal{U}([0, B]) \end{cases}$
- Sorting the τ_i into $\tau_{(1)} \leq \dots \leq \tau_{(P)}$ and simulating the bridge sequentially, each point given the previous one and Y_T :

$$\left(Y_{\tau_{(i)}} \mid Y_{\tau_{(i-1)}}, Y_T \right) \sim \mathcal{N}(m_i, \sigma_i)$$

where $m_i = Y_{\tau_{(i-1)}} + \left(Y_T - Y_{\tau_{(i-1)}} \right) \frac{\tau_{(i)} - \tau_{(i-1)}}{T - \tau_{(i-1)}}$ and $\sigma_i = \frac{(\tau_{(i)} - \tau_{(i-1)})(T - \tau_{(i)})}{T - \tau_{(i-1)}}$

- Computing $\mathbb{1}_{\{Z=0\}} = \mathbb{1}_{\{\forall i, -c(Y_{\tau_i}) \leq V_i\}}$, whose conditional expectation given Y

is exactly $\exp\left(\int_0^T c(Y_s) ds\right)$

- Multiply the three highlighted terms

By repeating the operation and taking the mean and the standard deviation, our Monte-Carlo becomes an unbiased estimator of $E[f(X)]$.

b) Results on Black-Scholes with European call

First, we tested our code on the Black-Scholes model under European call payoff, to compare Beskos-Roberts with the exact value, that we can compute. Here are the results.

Parameters				True Price	M-C results		Test results
r	volatility	K	M	μ	$\hat{\mu}_{BR}$	$\hat{\sigma}_{BR}$	Passed (P)/ Failed (F)
0.01	0.1	100	10^3	4.5303	4.5511	0.2256	P
			10^5		4.5412	0.0222	P
		120	10^3	0.1900	0.2072	0.0506	P
			10^5		0.1895	0.0044	P
	0.3	100	10^3	12.4926	12.4639	0.6016	P
			10^5		12.5213	0.0595	P
		120	10^3	5.7718	5.7494	0.4353	P
			10^5		5.8014	0.0423	P
	0.5	100	10^3	20.3469	20.3115	0.9988	P
			10^5		20.3827	0.0986	P
		120	10^3	13.5696	13.5266	0.8593	P
			10^5		13.6071	0.0842	P
0.05	0.1	100	10^3	7.1539	7.5519	0.5442	P
			10^5		7.1806	0.0486	P
		120	10^3	0.4862	0.6194	0.1889	P
			10^5		0.4845	0.0138	P
	0.3	100	10^3	14.9609	15.0453	0.7845	P
			10^5		14.9972	0.0764	P
		120	10^3	7.2580	7.3361	0.5877	P
			10^5		7.2918	0.0561	P
	0.5	100	10^3	22.9099	22.9451	1.1742	P
			10^5		22.9512	0.1148	P
		120	10^3	15.5413	15.5494	1.0242	P
			10^5		15.5851	0.0991	P
0.1	0.1	100	10^3	11.3923	11.5734	2.1042	P
			10^5		11.3635	0.1421	P
		120	10^3	1.3276	1.8545	1.0787	P
			10^5		1.2893	0.0529	P
	0.3	100	10^3	18.4941	18.5152	1.1120	P
			10^5		18.5441	0.1064	P
		120	10^3	9.5114	9.5763	0.8648	P
			10^5		9.5528	0.0807	P
	0.5	100	10^3	26.4431	26.6748	1.4475	P
			10^5		26.5083	0.1399	P
		120	10^3	18.3132	18.4758	1.2794	P
			10^5		18.3765	0.1223	P

Table 4: **Beskos-Roberts results on the European call**

Out of the 36 possible combinations of parameters, no failure occurred.

c) Barrier option

Now we consider the new payoff :

$$\tilde{f}((X_t)_t) = (X_T - K)_+ \mathbb{1}_{\{\min_{s \leq T} X_s \geq L\}} = f((X_t)_t) \cdot \mathbb{1}_{\{\min_{s \leq T} X_s \geq L\}}$$

where L is a given threshold and f is the previous payoff (European call). This payoff is called the **barrier option**. The term $\mathbb{1}_{\{\min_{s \leq T} X_s \geq L\}}$ needs a specific treatment to be computed. To do so, we take interest into the function q given by :

$$q(t, s, a, b) = \mathbb{Q}(\exists r \in [t, s] \text{ such that } X_r < L \mid X_t = a, X_s = b)$$

The link between q and $\mathbb{1}_{\{\min_{s \leq T} X_s \geq L\}}$ comes from the fact that q verifies :

$$q(t, s, a, b) = \mathbb{Q}\left(\min_{r \in [t, s]} X_r < L \mid X_t = a, X_s = b\right)$$

The goal of the 6 following steps is to compute q . The 7th focuses on code.

1. The reflection principle :

Let us consider a standard Brownian motion W on $[0, t]$. By the **reflection principle** [12], we have, for any $m \geq w$:

$$\mathbb{P}\left(\max_{s \leq t} W_s \geq m, W_t \leq w\right) = \mathbb{P}(W_t \geq 2m - w) \quad (*)$$

Here is a graph to understand the principle :

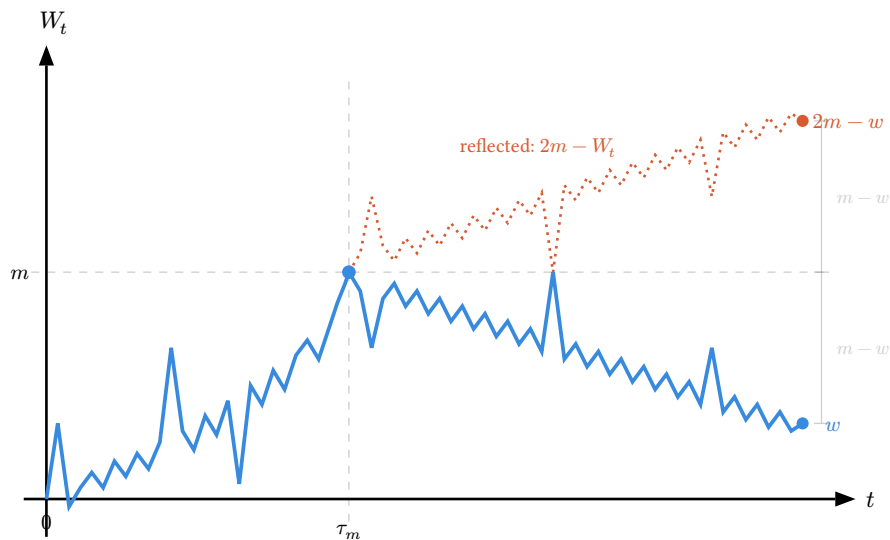


Figure 3: Reflection principle: the path after τ_m and its mirror image about $y = m$.

This principle allows us, thanks to the fact that W and $2m - W$ follow the same law on $[\tau_m, t]$, to get a new expression for the probability of the maximum of a Brownian motion intersected with an end-point condition.

2. Forcing the minimum to appear

We are interested in the $\min_{s \leq t} X_s$ more than in its maximum, but as :

$$\begin{aligned} \mathbb{P}\left(\min_{s \leq t} W_s \leq m, W_t \geq w\right) &= \mathbb{P}\left(-\max_{s \leq t}(-W_s) \leq m, -W_t \leq -w\right) \\ &= \mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m, -W_t \leq -w\right) \end{aligned}$$

We fall within the scope of the *reflection theorem*, since $-W$ is also a standard Brownian motion.

3. Conditional probability

We need to convert this intersection probability into a conditional one, to make a Brownian bridge appear. Thus :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t \geq w\right) = \frac{\mathbb{P}\left(\min_{s \leq t} W_s \leq m, W_t \geq w\right)}{\mathbb{P}(W_t \geq w)}$$

4. Converting the inequality conditions into equality ones

We do not really know how to deal with these “ $\mid W_t \geq w$ ”, what we do know is the *Brownian bridge*, that needs equality conditions instead of inequalities. Let's then take $\varepsilon > 0$. The idea is that we consider inequality conditions on very short intervals around w that shrink as $\varepsilon \rightarrow 0$:

$$\begin{aligned}
\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t \in [w, w + \varepsilon]\right) &= \mathbb{P}\left(-\max_{s \leq t}(-W_s) \leq m \mid -W_t \in [-w - \varepsilon, -w]\right) \\
&= \mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m \mid -W_t \in [-w - \varepsilon, -w]\right) \\
&= \frac{\mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m, -W_t \in [-w - \varepsilon, -w]\right)}{\mathbb{P}(-W_t \in [-w - \varepsilon, -w])} \\
&= \frac{G_t(-w) - G_t(-w - \varepsilon)}{\Phi_{-W_t}(-w) - \Phi_{-W_t}(-w - \varepsilon)} \\
&\text{where } \begin{cases} G_t(x) = \mathbb{P}\left(\max_{s \leq t}(-W_s) \geq -m, -W_t \leq x\right) \\ \Phi_{-W_t}(x) = \mathbb{P}(-W_t \leq x) = \Phi_{W_t}(\text{symmetric}) \end{cases} \\
&= \frac{G_t(-w) - G_t(-w - \varepsilon)}{\varepsilon} \cdot \frac{\varepsilon}{\Phi_{W_t}(w + \varepsilon) - \Phi_{W_t}(w)}
\end{aligned}$$

Now, by $\varepsilon \rightarrow 0$ and (*), we deduce :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t = w\right) = \frac{f_t(2m - w)}{f_t(w)}$$

by the reflection theorem, where f_t is the density of $\mathcal{N}(0, t)$.

5. Computing the wanted probability

At last, as $f_t(2m - w)/f_t(w) = \exp(-\frac{2}{t}m(m - w))$, we can compute the probability we've been looking for :

$$\mathbb{P}\left(\min_{s \leq t} W_s \leq m \mid W_t = w\right) = e^{-\frac{2}{t}m(m-w)} \quad (**)$$

Note : this formula holds for $m \leq \min(w, 0)$, which is exactly what the reflection principle asked for ; outside of that range its right-hand side would exceed 1. This is why the code below has to treat separately the points that are already under the barrier.

6. Linking back to the initial problem

We now have the $(**)$ formula for any standard Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and we want

$$q(t, s, a, b) = \mathbb{Q} \left(\min_{r \in [t, s]} X_r < L \mid X_t = a, X_s = b \right)$$

We know that under $\mathbb{Q}$, Y is a Brownian motion starting at $g(x)$. Let's change our two formulae to express them in terms of Y and $\mathbb{Q}$ instead of X and $\mathbb{P}$:

$$q(t, s, a, b) = \mathbb{Q} \left(\min_{r \in [t, s]} Y_r \leq g(L) \mid Y_t = g(a), Y_s = g(b) \right)$$

which is true because g is strictly increasing. Now, $(W_r)_{r \in [0, t]}$ in the formula $(**)$ has to be a standard Brownian motion while $(Y_r)_{r \in [t, s]}$ is not, as it has a non necessarily zero starting point $g(a)$. Hence we shift the time and define $(\tilde{W}_r)_{r \in [0, s-t]} := (Y_{r+t} - g(a))_{r \in [0, s-t]}$ which is a standard Brownian motion on $[0, s-t]$.

This leads us to consider :

$$\begin{aligned} & \mathbb{Q} \left(\min_{r \leq s-t} Y_{r+t} - g(a) \leq m \mid Y_{s-t+t} = w + g(a) \right) \\ = & \mathbb{Q} \left(\min_{r \in [t, s]} Y_r \leq m + g(a) \mid Y_s = w + g(a) \right) \end{aligned}$$

At last, we need $\begin{cases} g(a) + m = g(L) \\ w + g(a) = g(b) \end{cases} \implies \begin{cases} m = g(L) - g(a) \\ w = g(b) - g(a) \end{cases}$, that we plug into our formulae to get :

$$q(t, s, a, b) = \exp \left(-\frac{2}{s-t} (g(L) - g(a))(g(L) - g(b)) \right)$$

The very same computation, run with an arbitrary level λ and with the two endpoints written directly in Y , gives the general form of q that we keep for the whole report :

$$\begin{aligned} q(t, s, y_1, y_2, \lambda) &:= \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \leq \lambda \mid Y_t = y_1, Y_s = y_2\right) \\ &= \exp\left(-\frac{2}{s-t}(\lambda - y_1)(\lambda - y_2)\right) \end{aligned}$$

The barrier case we just derived is then $q(t, s, g(a), g(b), g(L))$. From now on, q always denotes this five-argument function.

7. Code

Now we know the exact value of

$$q(t, s, y_1, y_2, g(L)) = 1 - \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \geq g(L) \mid Y_t = y_1, Y_s = y_2\right) \quad \forall t, s$$

Which implies that, given the $P \sim \mathcal{P}(TB)$ intermediate points (τ_i, V_i) that we compute for *Beskos-Roberts* :

$$\begin{aligned} &\mathbb{Q}\left(\min_{[0, T]} Y_r \geq g(L) \mid Y_0, Y_{\tau_1}, \dots, Y_T\right) \\ &= \prod_{i=1}^{P+1} \mathbb{Q}\left(\min_{[\tau_{i-1}, \tau_i]} Y_r \geq g(L) \mid Y_{\tau_{i-1}}, Y_{\tau_i}\right) \\ &= \prod_{i=1}^{P+1} \left[1 - q(\tau_{i-1}, \tau_i, Y_{\tau_{i-1}}, Y_{\tau_i}, g(L))\right] \\ &= \prod_{i=1}^{P+1} \left[1 - \exp\left(-\frac{2}{\tau_i - \tau_{i-1}}(g(L) - Y_{\tau_{i-1}})(g(L) - Y_{\tau_i})\right)\right] \end{aligned}$$

where we took $\tau_0 = 0, \tau_{P+1} = T$ by convention, the first equality being the Markov property of Y under $\mathbb{Q}$: given its values at the τ_i , the $P + 1$ bridges are independent. where we took $\tau_0 = 0, \tau_{P+1} = T$ by convention.

Thus :

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}}[\tilde{f}(X)] &= \mathbb{E}^{\mathbb{Q}}\left[\tilde{f}(g^{-1}(Y)) \cdot e^{h(Y_T)-h(Y_0)+\int_0^T c(Y_s)ds}\right] \\
&= \mathbb{E}^{\mathbb{Q}}\left[f(g^{-1}(Y)) \cdot \mathbb{1}_{\{\min_{[0,T]} Y_s \geq g(L)\}} \cdot e^{h(Y_T)-h(Y_0)+\int_0^T c(Y_s)ds}\right]
\end{aligned}$$

That we can compute by Monte-Carlo by the following method :

- Doing the same thing as in Beskos-Roberts method on European call for $f(g^{-1}(Y))$, $e^{h(Y_T)-h(Y_0)}$, $e^{\int_0^T c(Y_s)ds}$
- $\mathbb{1}_{\{\min_{[0,T]} Y_s \geq g(L)\}}$: we can compute

$$\prod_{i=1}^{P+1} \left[1 - \exp\left(-\frac{2}{\tau_i - \tau_{i-1}} (g(L) - Y_{\tau_{i-1}})(g(L) - Y_{\tau_i})\right) \right]$$

with the (τ_i, V_i) we have already simulated to get $e^{\int_0^T c(Y_s)ds}$.

Note : for the code, as soon as a τ_i verifies $Y_{\tau_i} \leq g(L)$, we set the indicator function to 0 without further computations.

d) Tests for the barrier call

1. European > Barrier

The first test we ran was just meant to check a property that would have allowed us to instantly detect an issue with the method if violated. That is the fact that, as $M \rightarrow \infty$, we expect the price of the *European call* to be greater than the price of the *barrier call*. Of course, for a few paths simulated (low M), we could observe the opposite. But for $M = 10^6$, we start to get that property regardless of the other parameters. Thus, we **pass** this test.

2. Price decreasing in L and converging to the European as $L \rightarrow 0$

The second test we ran consists of the next plot :

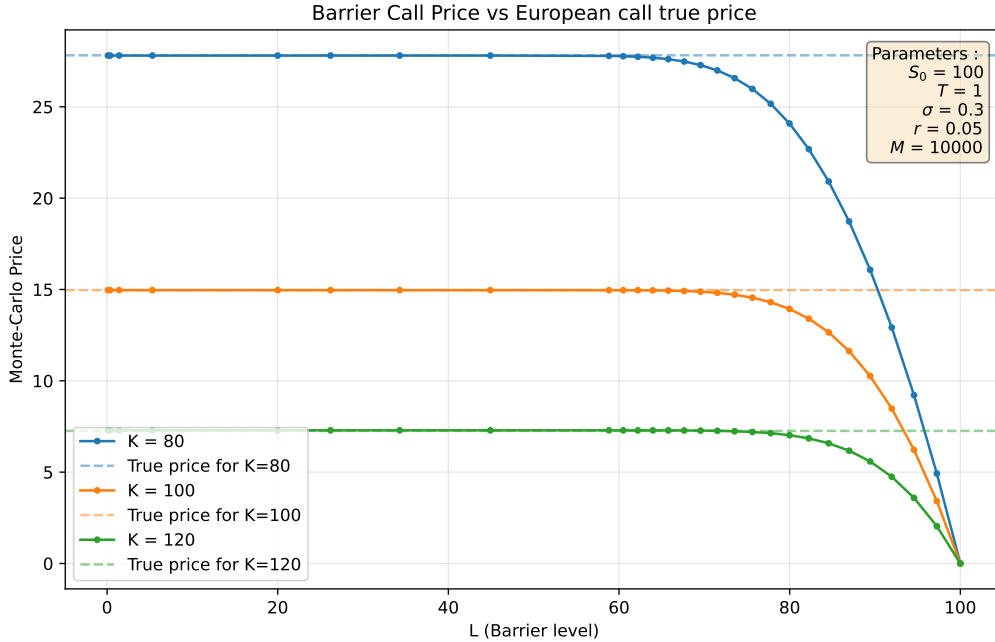


Figure 4: Barrier call convergence and monotony

Here is what we can say about it :

- There is a plateau at the European price for low L values. This is the convergence we were expecting : the lower the barrier, the more (X_t) paths reach $t = T$ without hitting it. This is exactly the European payoff behavior, hence the proximity. For $K = 80$, the plateau is reached as soon as $L \simeq 60$, but for $K = 120$, it is reached at $L \simeq 80$. This is because for low K , the paths live closer to the barrier, so more of them hit it and disconnect from the plateau. But for high K , the only surviving payoffs are the higher ones, which evolved far above the barrier, having less chance of hitting it.
- The previous point has for consequence that you can approximate the problem of pricing a barrier call by pricing the equivalent European call, for low L values. This “low” criterion **depends on σ and T** : the higher the volatility, the higher the probability that the path hits the barrier and disconnect from the European and the longer T the more opportunities (X_t) has to fall under the barrier too.
- The price is decreasing in L as expected. This is because the higher L the more paths hit it, and the lower L the bigger the surviving

area. For L close to S_0 , we expect almost all paths (that start at S_0) to hit the barrier, ending up in a payoff of 0.

- This test is qualitative : it gives us a trend that we compare to what we expected, but it doesn't allow us to assess that our method works nor that the code is correct.

3. Quantitative test

This last test checks that this q computing method, combined with Beskos-Roberts, is actually an unbiased method that computes $\mathbb{E}^{\mathbb{P}}[\tilde{f}((X_t)_t)]$ for $\tilde{f}$ the barrier payoff. To do so, we adapted the Euler scheme we first used for the European call to the barrier call. We expect it to be a biased estimator of the barrier call price that overestimates it. The reason is that the Euler scheme discretizes $[0, T]$ into a grid $\mathcal{G} = \{0 + i\frac{T}{N}, i \in \llbracket 0, N \rrbracket\}$ that might miss a point $\tau \notin \mathcal{G}$ such that $X_\tau < L$. The discretized path $\tilde{X}^N$ might verify $\min_{t \leq T} \tilde{X}_t^N \geq L$ while the actual path X has $\min_{t \leq T} X < L$. Hence the overestimation :

$$\mathbb{E}^{\mathbb{P}}[\tilde{f}((\tilde{X}_{t_i}^N)_{t_i \in \mathcal{G}})] \geq \mathbb{E}^{\mathbb{P}}[\tilde{f}((X_t)_t)]$$

As an estimator, we also expect from $\tilde{X}$ that :

$$\mathbb{E}^{\mathbb{P}}[\tilde{f}((\tilde{X}^N))] \xrightarrow{N \rightarrow \infty} \mathbb{E}^{\mathbb{P}}[\tilde{f}(X)]$$

Finally, the last thing our test has to show is the $O(1/\sqrt{N})$ convergence rate of the Euler scheme : monitoring the barrier on the grid only makes the weak order fall from 1 to 1/2, a fact we take for granted here [8] and only check numerically. Here is the result of the test, where we draw the curves in $\frac{1}{\sqrt{N}}$ instead of N to identify the convergence rate instantly :

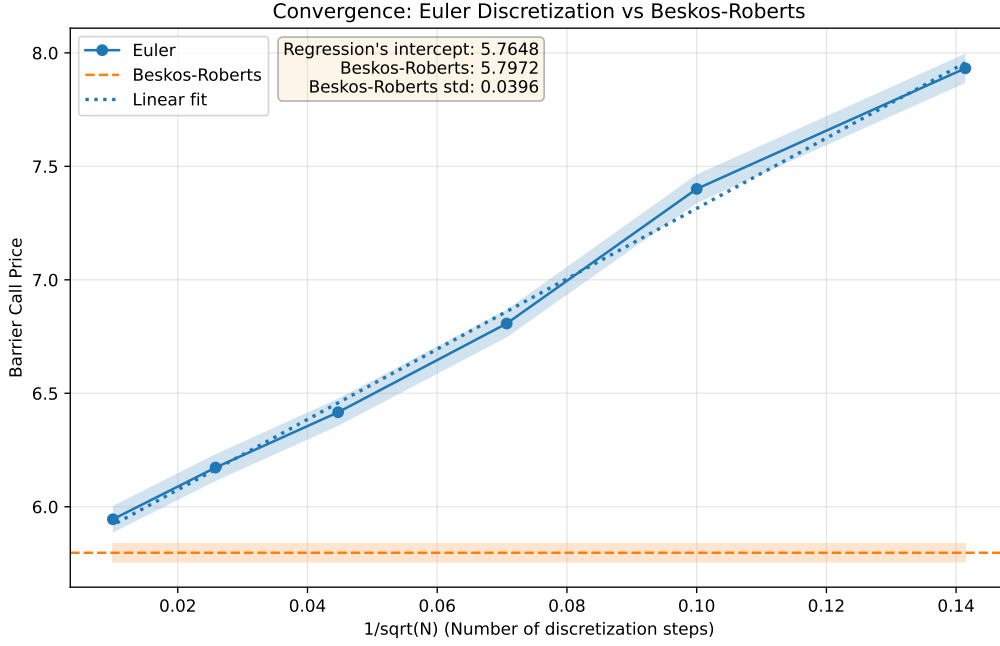


Figure 5: Euler scheme converging to barrier call unbiased method.

As expected, the Euler scheme overestimates the true price as it is decreasing. The linear regression in $1/\sqrt{N}$ shows that the Euler scheme converges :

- with a $O(1/\sqrt{N})$ rate
- toward the Beskos-Roberts price : intercept $\in [\mu_{BR} \pm \sigma_{BR}]$

Thus, Beskos-Roberts is an unbiased estimator of the barrier call price.

The method is **approved** by these 3 tests.

V Conclusion

Starting from a general one-dimensional diffusion, we built an estimator of $\mathbb{E}^{\mathbb{P}}[f(X)]$ that carries **no discretisation bias at all**. The road went through three steps : the *Lamperti* transform $Y = g(X)$, which makes the volatility constant ; *Girsanov-Maruyama*, which removes the drift and turns Y into a Brownian motion under $\mathbb{Q}$; and a last use of Itô formula, which trades the stochastic integral for the deterministic $\int_0^T c(Y_s)ds$. From there, *Beskos-Roberts* [1] replaces the exponential of that integral by a rejection experiment – a Poisson variable that we only need to see equal to zero – so that no time step ever appears : there are only random variables to simulate. The barrier case then asked for one more

ingredient, the probability q for a bridge to cross a level, which the reflection principle gives in closed form.

This method was not only derived, but **tested**. The Euler scheme, validated first on its own against the exact Black-Scholes price, served as the reference against which Beskos-Roberts was checked : on the European call, where the two agree over the whole grid of parameters, and on the barrier call, where the extrapolated $N \rightarrow \infty$ limit of the biased scheme and the unbiased estimate agree within one standard deviation.

Two limitations are worth keeping in mind for what follows. The construction asks $-c$ to be non-negative and bounded, which is a genuine restriction, and the reason why **Chapter III** exists. And it says nothing about the **variance** of the estimator, which is the quantity that decides how expensive the computation really is – the object of **Chapter II**.

Chapter II

We now consider the new CEV model given by :

$$(\mathbf{CEV}) : \begin{cases} dX_t = \sigma_0 X_t^\beta dW_t \\ X_0 = x \end{cases}$$

where $\sigma_0 > 0$ and $\beta \in]0, 1]$ [10]. If $\beta = 1$, the model corresponds to Black-Scholes :

$$(\mathbf{BS}) : \begin{cases} dX_t^{\mathbf{BS}} = \sigma_0 X_t^{\mathbf{BS}} dW_t \\ X_0^{\mathbf{BS}} = x \end{cases}$$

I Introduction

The main purpose of this second chapter is to build a new estimator for $\mathbb{E}^{\mathbb{P}}[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}}]$ where X follows **(CEV)**, $L \in \mathbb{R}$ is the barrier and x , the starting point, is assumed to be greater than L . We will use a lot of results from Chapter I on **(BS)**.

If we define the previous barrier European call price under Black-Scholes by :

$$C^{\mathbf{BS}}(T, x) = \mathbb{E}^{\mathbb{P}}[(X_T^{\mathbf{BS}} - K)_+ \cdot \mathbb{1}_{\{\min_{0 \leq t \leq T} X_t^{\mathbf{BS}} > L\}}]$$

we need to show the following equality :

$$\begin{aligned} (\mathbf{E}) : & \mathbb{E}^{\mathbb{P}}[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}}] \\ &= C^{\mathbf{BS}}(T, x) + \int_0^T \mathbb{E}^{\mathbb{P}}[(\mathcal{L} - \mathcal{L}^{\mathbf{BS}})C^{\mathbf{BS}}(T - s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}}] ds \end{aligned}$$

where $\mathcal{L}$ and $\mathcal{L}^{\mathbf{BS}}$ are called the generators of **(CEV)** and **(BS)** respectively.

II $\mathcal{L}$ and $\mathcal{L}^{\text{BS}}$ generators computation

Let's start by computing the two generators. For the record :

Theorem : Generator

A generator is an operator $\mathcal{L} : f \rightarrow \mathcal{L}f$ defined as follows :

$$\frac{\mathbb{E}^{\mathbb{P}} \left[f \left(X_t^{(\mathbf{M})} \right) \right] - f(x)}{t} \xrightarrow[t \rightarrow 0]{} \mathcal{L}f$$

where $(X_t)_t$ is the solution of a given stochastic model $(\mathbf{M})$ of starting point x under $\mathbb{P}$. $\mathcal{L}$ is called the generator of $(\mathbf{M})$.

So, for **(CEV)**, we have by Itô :

$$\begin{aligned} df(X_t) &= f'(X_t)dX_t + \frac{1}{2}f''(X_t)\sigma^2(X_t)dt \\ &= f'(X_t)\sigma_0 X_t^\beta dW_t + \frac{1}{2}\sigma_0^2 X_t^{2\beta} f''(X_t)dt \\ \Rightarrow f(X_t) &= f(x) + \int_0^t f'(X_s)\sigma_0 X_s^\beta dW_s + \frac{1}{2} \int_0^t \sigma_0^2 X_s^{2\beta} f''(X_s)ds \end{aligned}$$

We assume from now on that f is regular enough for

$$\mathbb{E}^{\mathbb{P}} \left[\int_0^t (f'(X_s)\sigma_0 X_s^\beta)^2 ds \right] < +\infty, \quad \forall t$$

to hold – it is for instance the case when f' has polynomial growth, since $\beta \leq 1$ keeps the moments of X under control. Under this assumption, $\left(\int_0^t f'(X_s)\sigma_0 X_s^\beta dW_s \right)_{t \in [0, T]}$ is a martingale. Consequently, we have :

$$\mathbb{E}^{\mathbb{P}} [f(X_t)] = f(x) + \frac{1}{2} \int_0^t \sigma_0^2 \mathbb{E}^{\mathbb{P}} [X_s^{2\beta} f''(X_s)] ds$$

Note : to exchange the $\int$ and $\mathbb{E}$ signs, we need (Fubini) f to verify $\mathbb{E}^{\mathbb{P}} \left[\int_0^t |X_s^{2\beta} f''(X_s)| ds \right] < +\infty \quad \forall t$.

Under those conditions, we can now compute $\mathcal{L}f$ and $\mathcal{L}^{\text{BS}}f$ as :

$$\frac{\mathbb{E}^{\mathbb{P}}[f(X_t)] - f(x)}{t} = \frac{1}{t} \int_0^t \frac{1}{2} \sigma_0^2 \mathbb{E}^{\mathbb{P}}[X_s^{2\beta} f''(X_s)] ds \xrightarrow{t \rightarrow 0} \frac{1}{2} \sigma_0^2 x^{2\beta} f''(x)$$

So eventually :

$$\mathcal{L}f = \frac{1}{2} \sigma_0^2 x^{2\beta} f''(x)$$

and

$$\mathcal{L}^{\text{BS}}f = \frac{1}{2} \sigma_0^2 x^2 f''(x)$$

III Proving the formula

1 Starting point and ending point computation

Now that we now what $\mathcal{L}$ and $\mathcal{L}^{\text{BS}}$ are, we can derive the equation (E), the most important of this report. Let's start by defining the random process $(M_t)_t$ by : $\forall s \in [0, T], M_s := C^{\text{BS}}(T - s, X_s) \cdot \mathbb{1}_{\{\min_{t \in [0, s]} X_t > L\}}$

We can start by deriving two important values of this process :

- $M_0 = C^{\text{BS}}(T, X_0) \cdot \mathbb{1}_{\{\min_{t \in [0, 0]} X_t > L\}} = C^{\text{BS}}(T, x) \cdot \mathbb{1}_{\{x > L\}} = C^{\text{BS}}(T, x)$ as x is supposed to be greater than L .
- $M_T = C^{\text{BS}}(0, X_T) \cdot \mathbb{1}_{\{\min_{t \leq T} X_t > L\}} \stackrel{(*)}{=} \mathbb{E}^{X_T} \left[(X_T^{\text{BS}} - K)_+ \mathbb{1}_{\{\min_{t \in [0, 0]} X_t^{\text{BS}} > L\}} \right] \times \mathbb{1}_{\{\min_{t \in [0, T]} X_t > L\}} = (X_T - K)_+ \mathbb{1}_{\{\min_{t \leq T} X_t > L\}}$

(*) $\mathbb{E}^z[Z_t]$ is the expectation under the probability space where $Z_0 = z$.

With these two values of (M_s) , and using Itô's formula, we will prove (E).

2 The stopping time θ

In order to use Itô's formula on M_s as a function of s and X_s , we need to compute the first and second derivatives with respect to X_s of M_s . The thing is, M_s depends on $\min_{t \leq s} X_t$, which is not a function of X_s but a functional of the whole path up to s . Also, as M_s needs to be $C^{1,2}$, we can not consider it as a function of s and $\tilde{X}_s = (X_s, \min_{t \leq s} X_t)$ either, because we would lose the C^2 condition with respect to space. This is due to the fact that $\mathbb{1}_{\{\min_{t \leq s} X_t > L\}}$ is not even continuous with respect to $\tilde{X}_s$.

The trick here is to define

$$\theta = \inf\{t \in [0, T] : X_t \leq L\}$$

θ is a stopping time which verifies the fundamental property :

$$M_s = C^{\text{BS}}(T - s \wedge \theta, X_{s \wedge \theta})$$

. Here is the proof of that statement :

$$M_s = \mathbb{E}^{\mathbb{P}} \left[(X_{T-s}^{\text{BS}} - K)_+ \cdot \mathbb{1}_{\{\min_{t \leq T-s} X_t^{\text{BS}} > L\}} \right] \times \mathbb{1}_{\{\min_{t \leq s} X_t > L\}}$$

⚠ : $X_t \neq X_t^{\text{BS}}$, the first follows the (CEV) model.

$$\begin{aligned} &= \mathbb{E}^{\mathbb{P}} \left[(X_{T-s}^{\text{BS}} - K)_+ \cdot \mathbb{1}_{\{\min_{t \leq T-s} X_t^{\text{BS}} > L\}} \right] \times \mathbb{1}_{\{s < \theta\}} \\ &= \begin{cases} \mathbb{E}^{\mathbb{P}} \left[(X_{T-s}^{\text{BS}} - K)_+ \cdot \mathbb{1}_{\{\min_{t \leq T-s} X_t^{\text{BS}} > L\}} \right] & \text{if } s < \theta \\ 0 & \text{else} \end{cases} \\ &= \begin{cases} C^{\text{BS}}(T - s, X_s) & \text{if } s < \theta \\ 0 & \text{else} \end{cases} \end{aligned}$$

$$\begin{aligned} C^{\text{BS}}(T - s \wedge \theta, X_{s \wedge \theta}) &= \begin{cases} C^{\text{BS}}(T - s, X_s) & \text{if } s < \theta \\ C^{\text{BS}}(T - \theta, X_\theta) & \text{else} \end{cases} \\ &= \begin{cases} C^{\text{BS}}(T - s, X_s) & \text{if } s < \theta \\ C^{\text{BS}}(T - \theta, L) & \text{else} \end{cases} \quad \text{as } X. \text{ is continuous} \\ &= \begin{cases} C^{\text{BS}}(T - s, X_s) & \text{if } s < \theta \\ 0 & \text{else} \end{cases} \\ &= M_s \end{aligned}$$

This equation allows us to use Itô's formula on $C^{\text{BS}}(T - s, X_s)$ as a function of s and X_s . We will prove later that it is a $C^{1,2}$ function as needed to apply Itô.

3 Applying Itô

Let's use $F(t, X_t) := C^{\text{BS}}(T - t, X_t)$. By Itô, we have :

$$dF(t, X_t) = \partial_t F(t, X_t)dt + \partial_x F(t, X_t)dX_t + \frac{1}{2}\partial_{x^2}^2 F(t, X_t)d\langle X, X \rangle_t$$

In the following lines, we define $f(x) := (x - K)_+$ and we consider that $\mathbb{E}^\alpha[Z]$, for a given process $(Z_t)_t$, refers to the expectation in a probability space such that $Z_0 = \alpha$.

a) Computing $\partial_t F$

To compute $\partial_t F(t, X_t) = -\frac{dC^{\text{BS}}}{d\tau}(\tau, X_t)$ with $\tau := T - t$, let's have a look at :

$$\mathbb{E}^z \left[f(X_{\tau+h}^{\text{BS}}) \mathbb{1}_{\{\min_{[h, h+\tau]} X^{\text{BS}} > L\}} | \mathcal{F}_h \right] = \mathbb{E}^{X_h^{\text{BS}}} \left[f(X_\tau^{\text{BS}}) \mathbb{1}_{\{\min_{[0, \tau]} X^{\text{BS}} > L\}} \right]$$

because X^{BS} is a Markov process as a solution of (BS)

$$= C^{\text{BS}}(\tau, X_h^{\text{BS}})$$

$$\begin{aligned} \text{So } \mathbb{E}^z \left[f(X_{\tau+h}^{\text{BS}}) \mathbb{1}_{\{\min_{[h, h+\tau]} X^{\text{BS}} > L\}} \right] &= \mathbb{E}^z \left(\mathbb{E}^z \left[f(X_{\tau+h}^{\text{BS}}) \mathbb{1}_{\{\min_{[h, h+\tau]} X^{\text{BS}} > L\}} | \mathcal{F}_h \right] \right) \\ &= \mathbb{E}^z [C^{\text{BS}}(\tau, X_h^{\text{BS}})] \quad \forall z \in \mathbb{R}, h > 0. \end{aligned}$$

Thanks to this, we can now easily compute $\partial_t F$, since :

$$\begin{aligned} &\frac{C^{\text{BS}}(\tau + h, z) - C^{\text{BS}}(\tau, z)}{h} \\ &= \frac{\mathbb{E}^z \left[f(X_{\tau+h}^{\text{BS}}) \mathbb{1}_{\{\min_{[h, h+\tau]} X^{\text{BS}} > L\}} \right] - \mathbb{E}^z \left[f(X_\tau^{\text{BS}}) \mathbb{1}_{\{\min_{[0, \tau]} X^{\text{BS}} > L\}} \right]}{h} \\ &= \frac{\mathbb{E}^z [C^{\text{BS}}(\tau, X_h^{\text{BS}})] - C^{\text{BS}}(\tau, z)}{h} \end{aligned}$$

We recognize the definition of the $\mathcal{L}^{\text{BS}}$ generator

$$\xrightarrow{h \rightarrow 0} \mathcal{L}^{\text{BS}}(C^{\text{BS}}(\tau, \cdot))[z]$$

So we just proved that $\frac{d}{d\tau} C^{\text{BS}}(\tau, X_t) = \frac{1}{2}\sigma_0^2 X_t^2 \partial_{x^2}^2 C^{\text{BS}}(\tau, X_t)$, hence :

$$\frac{\partial F}{\partial t}(t, X_t) = -\frac{1}{2}\sigma_0^2 X_t^2 \frac{\partial^2}{\partial x^2} C^{\text{BS}}(T - t, X_t)$$

which is the first component we need to write Itô's formula properly.

b) Computing the brackets

This part is easier than the previous one :

$$\frac{1}{2}\partial_{x^2}^2 F(t, X_t)d\langle X, X \rangle_t = \frac{1}{2}\partial_{x^2}^2 C^{\text{BS}}(T-t, X_t)\sigma_0^2 X_t^{2\beta} dt$$

$$\text{since } dX_t = \sigma_0 X_t^\beta dW_t$$

c) Gathering everything

When merging the previous results, we can compute $dF(t, X_t)$:

$$\begin{aligned} dF(t, X_t) &= \left[\frac{1}{2}\sigma_0^2 (X_t^{2\beta} - X_t^2) \frac{\partial^2}{\partial x^2} C^{\text{BS}}(T-t, X_t) \right] dt \\ &\quad + \frac{\partial F}{\partial x}(t, X_t) dX_t \\ &= (\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T-t, X_t)dt + \frac{\partial F}{\partial x}(t, X_t)\sigma_0 X_t^\beta dW_t \end{aligned}$$

So :

$$\begin{aligned} dC^{\text{BS}}(T-s, X_s) &= (\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T-s)ds \\ &\quad + \sigma_0 X_s^\beta \partial_x C^{\text{BS}}(T-s, X_s) dW_s \end{aligned}$$

By integrating this equation, we get the integrated form :

$$\begin{aligned}
\forall t \in [0, T], \quad C^{\text{BS}}(T - t, X_t) &= C^{\text{BS}}(T, x) \\
&+ \int_0^t (\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T - s, X_s) ds \\
&+ \int_0^t \partial_x C^{\text{BS}}(T - s, X_s) \sigma_0 X_s^\beta dW_s \\
\Rightarrow (M_t =) \quad C^{\text{BS}}(T - t \wedge \theta, X_{t \wedge \theta}) &= C^{\text{BS}}(T, x) \\
&+ \int_0^{t \wedge \theta} (\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T - s, X_s) ds \\
&+ \int_0^{t \wedge \theta} \partial_x C^{\text{BS}}(T - s, X_s) \sigma_0 X_s^\beta dW_s \\
&= C^{\text{BS}}(T, x) \\
&+ \int_0^t \mathbb{1}_{\{s \leq \theta\}} (\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T - s, X_s) ds \\
&+ \int_0^t \mathbb{1}_{\{s \leq \theta\}} \sigma_0 X_s^\beta \partial_x C^{\text{BS}}(T - s, X_s) dW_s
\end{aligned}$$

d) Taking the expectation

At last, we can take the expectation in the last equation and use the fact that $\mathbb{1}_{\{s \leq \theta\}} = \mathbb{1}_{\{\min_s X > L\}}$ to state :

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}}[M_t] &= C^{\text{BS}}(T, x) + \mathbb{E}^{\mathbb{P}} \left[\int_0^t (\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T - s, X_s) \mathbb{1}_{\{\min_s X > L\}} ds \right] \\
&+ \mathbb{E}^{\mathbb{P}} \left[\int_0^t \mathbb{1}_{\{\min_s X > L\}} \sigma_0 X_s^\beta \partial_x C^{\text{BS}}(T - s, X_s) dW_s \right]
\end{aligned}$$

One last step to make this strictly equivalent to **(E)** is to prove that the last expectation is 0. This is due to $\left(\int_0^t \mathbb{1}_{\{\min_s X > L\}} \sigma_0 X_s^\beta \partial_x C^{\text{BS}}(T - s, X_s) dW_s \right)_t$ being a martingale of starting point 0. We won't give a proof of that here, but the only thing to show is that :

$$\forall t, \quad \mathbb{E}^{\mathbb{P}} \left[\int_0^t \mathbb{1}_{\{\min_s X > L\}} \sigma_0^2 X_s^{2\beta} (\partial_x C^{\text{BS}}(T - s, X_s))^2 ds \right] < +\infty$$

e) Conclusion

Finally, using the derivation of M_T we did in Section 1 and assuming $\left(\int_0^t \mathbb{1}_{\{\min_s X > L\}} \sigma_0 X_s^\beta \partial_x C^{\text{BS}}(T-s, X_s) dW_s\right)_t$ is a martingale, plugging those two results in the last equation gives us, as wanted :

$$\begin{aligned} & \mathbb{E}^\mathbb{P} \left[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}} \right] = C^{\text{BS}}(T, x) \\ \text{(E)} \quad & + \int_0^T \mathbb{E}^\mathbb{P} \left[(\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T-s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}} \right] ds \end{aligned}$$

Now that we've proved the important equation, we will derive a formula of $(\mathcal{L} - \mathcal{L}^{\text{BS}}) C^{\text{BS}}(T-s, X_s)$, because it is the only challenge that remains before being able to use the equation in numerical computation. It will also allow us to understand why (E) does imply a reduction of variance, compared to the method presented in the first chapter. The derivation is very long, that's why it is the main point of the next 3 parts.

IV Clarifying C^{BS}

Thanks to the previous part, we know that we need C^{BS} to be twice differentiable, but is it the case ? The point of this part is to give a proper form to $C^{\text{BS}}(\tau, x)$.

Let's start :

$$\begin{aligned} C^{\text{BS}}(\tau, x) &= \mathbb{E}^\mathbb{P} \left[(X_\tau^{\text{BS}} - K)_+ \mathbb{1}_{\{\min_{0 \leq t \leq \tau} X_t^{\text{BS}} > L\}} \right] \\ &= \int_0^{+\infty} \int_0^{+\infty} (y - K)_+ \mathbb{1}_{\{l > L\}} p_\tau^x(y, l) dy dl \end{aligned}$$

(where $p_\tau^x(\cdot, \cdot)$ is the joint density of $(X_\tau, \min_{t \leq \tau} X_t)$ conditioned to $X_0 = x$)

$$\begin{aligned} &= \int_L^{+\infty} \int_K^{+\infty} (y - K) p_\tau^x(y, l) dy dl \\ &= \int_K^{+\infty} (y - K) \left(\int_L^{+\infty} p_\tau^x(y, l) dl \right) dy \\ &:= \int_K^{+\infty} (y - K) p_\tau^x(y) dy \end{aligned}$$

where we define $p_\tau^x(\cdot)$ as the restriction of $X_\tau|X_0=x$ density on $\{\min_{0 \leq t \leq \tau} X_t > L\}$. Note that $p_\tau^{x(\cdot)}$ is not an actual density as $\int_{\mathbb{R}} p_\tau^x < 1$. p_τ^x being the only x dependent function in C^{BS} , differentiating C^{BS} with respect to x requires us to compute p_τ^x , which is not an easy task. On the other hand, we proved in chapter I that a change of variable and probability can give us a new process Y that is a Brownian motion. That is the possibility we explore in next part.

V Using Y instead of X

A lot of the results used in this part are proved in Chapter I, particularly for the Lamperti transform $Y = g(X)$ and for the change of probability $\mathbb{P} \longleftrightarrow \mathbb{Q}$ by Girsanov. These two transformations allow us to do the next derivation :

$$\begin{aligned} C^{\text{BS}}(\tau, x) &= \mathbb{E}^{\mathbb{P}} \left[(X_\tau^{\text{BS}} - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} X_t^{\text{BS}} > L\}} \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[(g^{-1}(Y_\tau) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[(g^{-1}(Y_\tau) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \cdot \frac{d\mathbb{P}}{d\mathbb{Q}} \right] \end{aligned}$$

$$\text{with } \frac{d\mathbb{P}}{d\mathbb{Q}} = \frac{1}{L_\tau} \text{ (see chapter I for } L_t \text{ definition)}$$

$$\begin{aligned} &= \exp \left(\int_0^\tau a(Y_s) dW_s + \frac{1}{2} \int_0^\tau a^2(Y_s) ds \right) \\ &= e^{-\frac{1}{2}\sigma_0 W_\tau + \frac{1}{8}\sigma_0^2 \tau} \text{ since } a(y) = -\frac{1}{2}\sigma_0 \\ &= e^{-\frac{1}{2}\sigma_0(Y_\tau - g(x)) - \frac{1}{8}\sigma_0^2 \tau} \text{ as } dY_t = -\frac{1}{2}\sigma_0 dt + dW_t \\ &:= w(Y_\tau, \tau) \end{aligned}$$

$$\begin{aligned} \text{So } C^{\text{BS}}(\tau, x) &= \mathbb{E}^{\mathbb{Q}} \left[(g^{-1}(Y_\tau) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \cdot w(Y_\tau, \tau) \right] \\ &= \int_{g(K)}^{+\infty} \int_{g(L)}^{+\infty} (g^{-1}(u) - K) w(u, \tau) q_\tau^x(u, l) dl du \end{aligned}$$

where we define $q_\tau^x(\cdot, \cdot)$ as the joint density of $(Y_\tau, \min_{t \leq \tau} Y_t)$ conditioned to $Y_0 = g(x)$. This time, the density is much easier to compute. We know from Chapter I that :

$$q(t, s, y_1, y_2, \lambda) = \mathbb{Q}\left(\min_{r \in [t, s]} Y_r \leq \lambda | Y_t = y_1, Y_s = y_2\right)$$

which is the cumulative distribution function of $(\min_r Y | Y_t, Y_s)$

$$\Rightarrow \frac{\partial q}{\partial \lambda}(t, s, y_1, y_2, \lambda) = \tilde{q}(\lambda)$$

with $\tilde{q}$ the density of $(\min_r Y | Y_t, Y_s)$

$$\Rightarrow f_s(y_2) \cdot \frac{\partial q}{\partial \lambda}(t, s, y_1, y_2, g(L)) = \tilde{q}(g(L)) \cdot f_s(y_2)$$

If f_s is defined as the density of Y_s , $\tilde{q}(g(L)) \cdot f_s(y_2)$ is exactly the density of $(\min_r Y_r, Y_s)$ conditionally to $Y_t = y_1$, which is $q_\tau^x(\cdot, \cdot)$. Consequently :

$$\begin{aligned} q_\tau^x(u, l) &= f_{\tau(u)} \cdot \frac{\partial}{\partial \lambda} q(0, \tau, g(x), u, l) \text{ valid for } l \leq \min(g(x), u) \\ &= \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \frac{d}{dl} \left(e^{-\frac{2}{\tau}(l-g(x))(l-u)} \right) \\ &= -\frac{2}{\tau} \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} (2l - u - g(x)) e^{-\frac{2}{\tau}(l-g(x))(l-u)} \end{aligned}$$

Similarly to $p_\tau^x(\cdot)$, we are more interested into $q_\tau^x(\cdot)$ than into $q_\tau^x(\cdot, \cdot)$. Let's derive $q_\tau^x(u)$:

$$\begin{aligned} \forall u \geq g(L), \quad q_\tau^x(u) &:= \int_{g(L)}^{\min(g(x), u)} q_\tau^x(u, l) dl \\ &= -\frac{2}{\tau} \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \int_{g(L)}^{\min(g(x), u)} (2l - u - g(x)) \cdot e^{-\frac{2}{\tau}(l-g(x))(l-u)} dl \\ &= \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \end{aligned}$$

To conclude, we have the 3 equalities :

$$w^x(u, \tau) = e^{-\frac{1}{2}\sigma_0(u-g(x)) - \frac{1}{8}\sigma_0^2\tau}$$

$$q_\tau^x(u) = \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)}\right) \quad \forall u \geq g(L)$$

$$C^{\text{BS}}(\tau, x) = \int_{\max\{g(K), g(L)\}}^{+\infty} (g^{-1}(u) - K) w^x(u, \tau) q_\tau^x(u) du$$

VI Computing $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\cdot, \cdot)$

Let's look at the equation we want to prove :

$$\begin{aligned} & \mathbb{E}^\mathbb{P} \left[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}} \right] \\ &= C^{\text{BS}}(T, x) + \int_0^T \mathbb{E}^\mathbb{P} \left[(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T-s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}} \right] ds \end{aligned}$$

We have to compute $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T-s, x)$, with

$$(\mathcal{L} - \mathcal{L}^{\text{BS}})f = \frac{1}{2}\sigma_0^2(x^{2\beta} - x^2)f''(x)$$

This yields that C^{BS} has to be twice differentiable. Given the form of w and q_τ^x found above, $x \mapsto w^x(u, \tau)q_\tau^x(u)$ is twice differentiable. Let's now differentiate it twice to assert that we can dominate the second derivative by a L^1 function and conclude on C^{BS} twice differentiability. We start by defining the following useful expressions :

$$U = \frac{(u-g(x))^2}{2\tau}$$

$$B = g(L) - u$$

$$A = 2g(L) - g(x) - u$$

$$C = u - g(x)$$

$$E = -\frac{2}{\tau}(g(L) - g(x))(g(L) - u)$$

1 First derivatives

- We start by $\partial_x q_\tau^x$:

$$\begin{aligned}
\frac{\partial q_\tau^x}{\partial x}(u) &= \frac{1}{\sqrt{2\pi\tau}} \left[\frac{\partial}{\partial x} \left(e^{-\frac{(u-g(x))^2}{2\tau}} \right) \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right. \\
&\quad \left. - e^{-\frac{(u-g(x))^2}{2\tau}} \cdot \frac{\partial}{\partial x} \left(e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right] \\
&= \frac{1}{\sqrt{2\pi\tau}} \left[g'(x) \frac{u-g(x)}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right. \\
&\quad \left. - \frac{2}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} g'(x) (g(L)-u) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right] \\
&= \frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \left(u-g(x) - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} (2g(L)-g(x)-u) \right) \\
&= \frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-U} (C - Ae^E)
\end{aligned}$$

- Then $\partial_x w^x(u, \tau)$:

$$\frac{\partial w}{\partial x}(u, \tau) = \frac{\sigma_0 g'(x)}{2} w^x(u, \tau) = \frac{w^x(u, \tau)}{2x} \text{ because here } g(x) = \frac{1}{\sigma_0} \ln(x)$$

2 Second derivatives

- $\partial_{x^2}^2 q_\tau^x$:

$$\begin{aligned}
&\triangleright \frac{\partial}{\partial x} \left(u - g(x) - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} (2g(L) - g(x) - u) \right) \\
&\quad = -g'(x) + g'(x) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \\
&\quad \quad - (2g(L) - g(x) - u) \frac{2g'(x)}{\tau} \times (g(L) - u) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \\
&\quad = g'(x) \left[-1 + e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \left(1 - \frac{2}{\tau} (2g(L) - g(x) - u)(g(L) - u) \right) \right] \\
&\quad = g'(x) \left[-1 + e^E \left(1 - \frac{2}{\tau} AB \right) \right] \\
&\triangleright \frac{\partial}{\partial x} \left(\frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{g''(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} + \frac{g'(x)}{\tau\sqrt{2\pi\tau}} \frac{g'(x)(u-g(x))}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} \\
&= \frac{1}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \left(g''(x) + \frac{(g'(x))^2(u-g(x))}{\tau} \right) \\
&= \frac{e^{-U}}{\tau\sqrt{2\pi\tau}} \left(g''(x) + \frac{(g'(x))^2 C}{\tau} \right)
\end{aligned}$$

- Finally, gathering the two lines above and using $g''(x) = -\sigma_0 g'(x)^2$, which holds here because $g(x) = \frac{1}{\sigma_0} \ln(x)$:

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} q_\tau^x(u) &= \frac{g'(x)^2 e^{-U}}{\tau\sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C \right. \\
&\quad \left. + e^E \left(1 - \frac{2}{\tau} AB - \frac{AC}{\tau} + \sigma_0 A \right) \right]
\end{aligned}$$

- $\partial_{x^2}^2 w$:

$$\begin{aligned}
\frac{\partial}{\partial x} \frac{w^x(u, \tau)}{2x} &= -\frac{1}{2x^2} w^x(u, \tau) + \frac{w^x(u, \tau)}{4x^2} \\
&= -\frac{w^x(u, \tau)}{4x^2}
\end{aligned}$$

3 $\partial_{x^2}^2[w^x(u, \tau)q_\tau^x(u)]$ computation

We now have all the necessary elements to compute $\partial_{x^2}^2[w^x(u, \tau)q_\tau^x(u)]$ with Leibniz :

$$\begin{aligned}
\frac{\partial^2}{\partial x^2}[w(\tau, x)q_\tau^x(u)] &= \left(\frac{\partial^2}{\partial x^2}q_\tau^x(u) \right) w^x(u, \tau) + 2 \left(\frac{\partial}{\partial x}q_\tau^x(u) \right) \left(\frac{\partial}{\partial x}w^x(u, \tau) \right) \\
&\quad + q_\tau^x(u) \left(\frac{\partial^2}{\partial x^2}w^x(u, \tau) \right) \\
&= \frac{g'(x)^2 e^{-U}}{\tau \sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{2}{\tau}AB - \frac{AC}{\tau} + \sigma_0 A \right) \right] \\
&\quad \times w^x(u, \tau) + \frac{g'(x)}{\tau \sqrt{2\pi\tau}} e^{-U} (C - Ae^E) w^x \frac{u, \tau}{x} \\
&\quad - \frac{e^{-U}}{\sqrt{2\pi\tau}} (1 - e^E) w^x \frac{u, \tau}{4x^2} \\
&= \frac{g'(x)^2 e^{-U}}{\tau \sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A}{\tau}(2B + C) + \sigma_0 A \right) \right] \\
&\quad \times w^x(u, \tau) + \frac{g'(x)}{\tau \sqrt{2\pi\tau}} e^{-U} (C - Ae^E) w^x \frac{u, \tau}{x} \\
&\quad - \frac{e^{-U}}{\sqrt{2\pi\tau}} (1 - e^E) w^x \frac{u, \tau}{4x^2} \\
&= \frac{w^x(u, \tau) e^{-U}}{\tau \sqrt{2\pi\tau}} \left\{ g'(x)^2 \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A}{\tau}(2B + C) + \sigma_0 A \right) \right] \right. \\
&\quad \left. + \frac{g'(x)}{x} (C - Ae^E) - \frac{\tau}{4x^2} (1 - e^E) \right\}
\end{aligned}$$

4 $C^{\text{BS}} \in C^2$ final proof

Let's bound $x \in [x_-, x_+]$ and look at $\partial_{x^2}^2(w(\tau, x)q_\tau^x(u))$:

- $w^x(u, \tau) = \exp\left(-\frac{1}{2}\sigma_0(u - g(x)) - \frac{1}{8}\sigma_0^2\tau\right) = O(e^{-\sigma_0 u/2})$
- A, B and C are $O(u)$
- $e^E = \exp\left(-\frac{2}{\tau}(g(L) - g(x))(g(L) - u)\right) \leq O(e^{\kappa u})$

where $\kappa = \frac{2}{\tau}(g(L) - g(x)) < 0$ because we assume (X_t) starts over the barrier.

So overall, the big brace factor is $O(u^2)$, and multiplying it by $O(e^{-(\sigma_0 u)/2})$, which is bounded, doesn't change the $O(u^2)$ domination. At last, if we define $d(u) := \text{dist}(u, [g(x_-), g(x_+)])$, we have :

$$e^{-U} = e^{-\frac{1}{2\tau}(u-g(x))^2} \leq \begin{cases} e^{-\frac{1}{2\tau}(u-g(x_-))^2} & \text{if } u < g(x_-) \\ 1 & \text{if } u \in [g(x_-), g(x_+)] \\ e^{-\frac{1}{2\tau}(u-g(x_+))^2} & \text{if } u > g(x_+) \end{cases} = e^{-\frac{d(u)^2}{2\tau}}$$

Finally, as the integrand of $C^{\text{BS}}(\tau, x)$ is $(e^{\sigma_0 u} - K)w^x(u, \tau)q_\tau^x(u)$, we have the domination :

$$\begin{aligned} |(e^{\sigma_0 u} - K)\partial_{x^2}^2[w^x(u, \tau)q_\tau^x(u)]| &\leq O(u^2)e^{\sigma_0 u}e^{-\frac{d(u)^2}{2\tau}} \\ &= O(u^2)e^{\sigma_0 d(u) - \frac{d(u)^2}{2\tau}} (*) \\ &= O(u^2)e^{-\frac{1}{2\tau}(d(u) - \sigma_0 \tau)^2 + \frac{\sigma_0^2 \tau}{2}} \\ &= O(u^2) \times \text{gaussian} \end{aligned}$$

which is a domination by a L^1 function independent of x : C^{BS} is twice differentiable on $[x_-, x_+]$, and since $0 < x_- < x_+$ were arbitrary, on the whole of $]0, +\infty[$.

Note :

- $(*)$: we have indeed $e^{\sigma_0 u} = O(1) \cdot e^{\sigma_0 d(u)}$ since we always have $u \leq d(u) + g(x_+)$.
- We have to show that the integrand of C^{BS} , and the first derivative of this integrand with respect to x are L^1 with respect to u . For the first one, it is implied by the positivity of that integrand and because C^{BS} is well defined. For the first derivative, we can show that

$$|(e^{\sigma_0 u} - K)\partial_x[w^x(u, \tau)q_\tau^x(u)]| \leq O(u) \times \text{gaussian}$$

which makes it L^1 too.

5 Final result

Here is where we are at :

$$\begin{aligned}
& (\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\tau, x) \\
&= \frac{1}{2}\sigma_0^2(x^{2\beta} - x^2) \int_{\max(g(K), g(L))}^{+\infty} (g^{-1}(u) - K) \\
&\times \frac{w^x(u, \tau)e^{-U}}{\tau\sqrt{2\pi\tau}} \left\{ g'(x)^2 \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A}{\tau}(2B + C) + \sigma_0 A \right) \right] \right. \\
&\left. + \frac{g'(x)}{x}(C - Ae^E) - \frac{\tau}{4x^2}(1 - e^E) \right\} du
\end{aligned}$$

This is a very big formula, we can simplify it :

- $w^x(u, \tau)e^{-U} = e^{-\frac{1}{2}\sigma_0(u-g(x)) - \frac{1}{8}\sigma_0^2\tau} e^{-\frac{(u-g(x))^2}{2\tau}} = e^{-\frac{1}{2\tau}(u-g(x)+\frac{\sigma_0\tau}{2})^2} = e^{-\frac{1}{2\tau}(u-\mu)^2}$ where we define $\mu := g(x) - \frac{\sigma_0\tau}{2}$ the mean of Y_τ under $\mathbb{P}$.
- As $g'(x)^2 = \frac{1}{\sigma_0^2 x^2}$ and $\frac{g'(x)}{x} = \frac{1}{\sigma_0 x^2}$, we can factor the biggest brace factor by $1/x^2$.
- $2B + C = 2g(L) - 2u + u - g(x) = A$, so $\frac{A}{\tau}(2B + C)$ becomes $\frac{A^2}{\tau}$

These 3 points simplifies our expression :

$$\begin{aligned}
& (\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(x, \tau) \\
&= \frac{1}{2}\sigma_0^2(x^{2(\beta-1)} - 1) \int_{\max(g(L), g(K))}^{+\infty} (g^{-1}(u) - K) \frac{e^{-\frac{1}{2\tau}(u-\mu)^2}}{\tau\sqrt{2\pi\tau}} \times \left\{ \frac{C - Ae^E}{\sigma_0} \right. \\
&\left. - \frac{\tau}{4}(1 - e^E) + \frac{\left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A^2}{\tau} + \sigma_0 A \right) \right]}{\sigma_0^2} \right\} du
\end{aligned}$$

Now here is what we notice, if we define $P(\xi) := \frac{1}{\tau}(\xi^2 - \sigma_0\tau\xi - \tau)$:

$$\begin{aligned}
& \left\{ \frac{C - Ae^E}{\sigma_0} - \frac{\tau}{4}(1 - e^E) + \frac{\left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A^2}{\tau} + \sigma_0 A \right) \right]}{\sigma_0^2} \right\} \\
&= \left\{ \frac{P(C) - P(A)e^E}{\sigma_0^2} + \frac{C - Ae^E}{\sigma_0} - \frac{\tau}{4}(1 - e^E) \right\} \\
&= \left\{ \left(\frac{P(C)}{\sigma_0^2} + \frac{C}{\sigma_0} - \frac{\tau}{4} \right) - \left(\frac{P(A)}{\sigma_0^2} + \frac{A}{\sigma_0} - \frac{\tau}{4} \right) e^E \right\} \\
&= \{R(C) - R(A)e^E\}
\end{aligned}$$

where $R(\xi) := \frac{P(\xi)}{\sigma_0^2} + \frac{\xi}{\sigma_0} - \frac{\tau}{4} = \frac{1}{\sigma_0^2} \left(\frac{\xi^2}{\tau} - 1 \right) - \frac{\tau}{4}$.

Finally,

$$\begin{aligned} & (\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\tau, x) \\ &= \frac{1}{2}\sigma_0^2(x^{2(\beta-1)} - 1) \int_M^{+\infty} f_K(u) \frac{e^{-\frac{(u-\mu)^2}{2\tau}}}{\tau\sqrt{2\pi\tau}} (R(C) - R(A)e^E) du \end{aligned}$$

where :

$$f_K : u \mapsto g^{-1}(u) - K$$

$$M := \max\{g(L), g(K)\}$$

$$A_x = 2g(L) - g(x) - u$$

$$C_x = u - g(x)$$

$$\mu_x := g(x) - \frac{\sigma_0\tau}{2}$$

$$E_x = -\frac{2}{\tau}(g(L) - g(x))(g(L) - u)$$

VII Qualitative intuition

Let us recall that this whole chapter focus on the reduction of variance [14, 15, 16] in the numerical computation of $\mathbb{E}^{\mathbb{P}}[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}}]$ where X is a stochastic process solution of the **(CEV)** model. So far, we proved the equation **(E)** given by :

$$\begin{aligned} & \mathbb{E}^{\mathbb{P}}[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}}] = C^{\text{BS}}(T, x) \\ \text{(E)} \quad & + \int_0^T \mathbb{E}^{\mathbb{P}}[(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}}] ds \end{aligned}$$

This equation relies on the formula of $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\tau, x)$ derived in the previous part. But why is this a variance reduction method ? It would be interesting to measure how much **(E)** allows a variance reduction compared to other computation methods. Sadly, having a definitive answer would require coding

this method in Python and we have been asked to move on to another problem, addressed in Chapter III, before having that occasion. The code has been assigned to another researcher.

However, it doesn't mean we can't qualitatively discuss **(E)** for an instant. The first thing we notice is the $(x^{2(\beta-1)} - 1)$ factor in front of $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\tau, x)$ formula. This factor is the only one depending on β in the entire formula, and is converging to 0 as β approaches 1. More accurately, by defining $\varepsilon_t := \frac{1}{2}\sigma_0^2(X_t^{2(\beta-1)} - 1)$, the formula factorises as :

$$(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - t, X_t) = \varepsilon_t \cdot I_{X_t}$$

where

$$I_{X_t} := \int_M^{+\infty} f_K(u) \frac{e^{-\frac{(u-\mu_{X_t})^2}{2(T-t)}}}{(T-t)\sqrt{2\pi(T-t)}} \Theta_t du$$

and

$$\Theta_t = R(C_{X_t}) - R(A_{X_t}) \exp(E_{X_t})$$

do not depend on β .

Additionally, we have :

$$\begin{aligned} \varepsilon_t &= \frac{1}{2}\sigma_0^2(X_t^{2(\beta-1)} - 1) = \frac{1}{2}\sigma_0^2(e^{2(\beta-1)\ln(X_t)} - 1) \\ &= \sigma_0^2(\beta - 1)\ln(X_t) + o(\beta - 1) \end{aligned}$$

Consequently, as long as $\mathbb{E}^{\mathbb{P}}[I_{X_t}^2 \ln^2(X_t)] < +\infty$ - the barrier $X_t > L > 0$ being what keeps $\ln(X_t)$ under control on the dangerous side - we have :

$$\begin{aligned} \text{Var}((\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - t, X_t)) &= \text{Var}(\varepsilon_t \cdot I_{X_t}) \\ &= O((\beta - 1)^2) \end{aligned}$$

VIII Conclusion

In conclusion, this method reduces the variance as β approaches one, which corresponds to the (CEV) model approaching (BS). On the other hand, it has the major weakness that the numerical computation of $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - t, X_t)$ appears to be a problem in itself, that might be settled using Beskos-Roberts as we did in Chapter I. However, this would need to prove that Beskos-Roberts doesn't need the function c to be bounded, something we have yet to prove ourselves.

Why this chapter stands apart. The reader will have noticed that this chapter shares almost nothing with the previous one, apart from the Lamperti transform and the change of probability : another model, another way of handling the barrier, another technique, and no Monte-Carlo at all. The reason is the history of the internship itself. The subject I had been given was the decomposition (E) – express the (CEV) barrier price as the Black-Scholes one plus a correction, and study the variance reduction it provides. *Beskos-Roberts* was not part of that plan : it is the route I took, on my own initiative, to answer the question of **Chapter I**, where another unbiased scheme was expected. Yamada-sensei therefore had me carry out the work that had been planned in the first place, which is this chapter, while he examined that idea on his side. He then judged it worth pursuing, and sent me back to it with a new question : can the boundedness of c be removed ? That question became **Chapter III**. This chapter is thus best read as the branch that was planned, and the two others as the branch that grew out of a detour.

This marks the end of Chapter II, that will hopefully be useful to those in charge of the code.

Chapter III

We consider $D \subset \mathbb{R}$ and the general SDE :

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad X_0 = x \in D$$

We also define the convenient $\tau_X := \inf\{t : X_t \notin D\}$, a real $\eta > 0$ and

$$\begin{cases} N \sim \mathcal{P}(\eta) \\ (U_i)_{i \geq 1} \text{ i.i.d. } \sim \mathcal{U}([0, 1]) \end{cases} \quad N, (U_i)_i \text{ and } W \text{ being independent}$$

I A new formula for $\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right]$

Let's start by noticing that the price of a payoff killed as soon as X leaves D is $\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right]$, by definition of τ_X . We now want to derive a new expression for this fair-price that would use N and the U_i .

In Chapter I, we have shown that

$$\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\int_0^T c(Y_s) ds} \right]$$

where $\tau_Y := \inf\{t : Y_t \notin g(D)\}$, $y := Y_0$, $\mathbb{Q}$ is a probability space defined in Chapter I and c is a function defined in Chapter I too

First, we would like to prove the following representation, in the spirit of the *Poisson estimator* [4, 5] :

$$\exp \left(\int_0^T c(Y_s) ds \right) = \mathbb{E}^{\mathbb{Q}} \left[e^{\eta} \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\} \middle| Y \right]$$

Indeed, using that N and the U_i are independent of Y for the first line below, and that the U_i are i.i.d. for the second :

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{Tu_i}) \right\}} \middle| Y \right] &= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot \mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} c(Y_{Tu_i}) \right\}} \middle| Y, N = n \right] \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \mathbb{E}^{\mathbb{Q}} \left[c(Y_{Tu_i}) \middle| Y, N = n \right] \right\}} \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \int_0^1 c(Y_{Tu_i}) du_i \right\}}
\end{aligned}$$

where we can apply the change of variable : $s_i = Tu_i$

$$\begin{aligned}
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot e^{\eta \prod_{i=1}^n \left\{ \frac{T}{\eta} \int_0^T \frac{1}{T} c(Y_{s_i}) ds_i \right\}} \\
&= \sum_{n=0}^{+\infty} \mathbb{Q}(N = n) \cdot \frac{e^{\eta}}{\eta^n} \cdot \left(\int_0^T c(Y_s) ds \right)^n \\
&= \sum_{n=0}^{+\infty} e^{-\eta} \frac{\eta^n}{n!} \cdot \frac{e^{\eta}}{\eta^n} \cdot \left(\int_0^T c(Y_s) ds \right)^n \\
&= \sum_{n=0}^{+\infty} \frac{1}{n!} \left(\int_0^T c(Y_s) ds \right)^n \\
&= \exp \left(\int_0^T c(Y_s) ds \right)
\end{aligned}$$

With that in mind, let's look again at our

$$\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbf{1}_{\tau_X > T} \right] = \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbf{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\int_0^T c(Y_s) ds} \right]$$

in which we can substitute our new $e^{\int_0^T c(Y_s) ds}$ expression :

$$\begin{aligned}
& \mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} \mathbb{E}^{\mathbb{Q}} \left[e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \mid Y \right] \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[\mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \mid Y \right] \right]
\end{aligned}$$

Because $Y \mapsto f(g^{-1}(Y_T))$, $Y \mapsto \mathbb{1}_{\tau_Y > T}$ and $Y \mapsto h(Y_T)$ are measurable

$$= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \right]$$

In conclusion, this formula concludes this first part :

$$\begin{aligned}
& \mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] \\
&= \mathbb{E}^{\mathbb{Q}} \left[f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^{\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}} \right]
\end{aligned}$$

II Some additional work

Let's separate two cases : $\{N = 0\}$ and $\{N \geq 1\}$. For the convenience, we define :

$$Z := f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}$$

In the previous equation, it gives :

$$\begin{aligned} & \mathbb{E}^{\mathbb{P}} [f(X_T) \mathbb{1}_{\tau_X > T}] \\ &= \mathbb{E}^{\mathbb{Q}} [Z] = \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N=0}] + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}] \\ &= \mathbb{E}^{\mathbb{Q}} [Z | N = 0] \cdot \mathbb{Q}(N = 0) + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}] \\ &= \mathbb{E}^{\mathbb{Q}} [f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)} e^\eta] \cdot e^{-\eta} \\ &\quad + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}] \\ &= \mathbb{E}^{\mathbb{Q}} [f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T) - h(y)}] \\ &\quad + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}] = C + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}] \end{aligned}$$

Let's dig deeper for C :

$$\begin{aligned} C &= \mathbb{E}^{\mathbb{Q}} [f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} e^{h(Y_T) - h(y)}] \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} f(g^{-1}(z)) \mathbb{1}_{\tau > T} e^{h(z) - h(y)} p(\tau, y, z) dz d\tau \end{aligned}$$

where p is the joint density of (Y_T, τ_Y) conditionally on $\{Y_0 = y\}$

$$\begin{aligned} &= \int_{\mathbb{R}} f(g^{-1}(z)) e^{h(z) - h(y)} \left(\int_{\mathbb{R}} \mathbb{1}_{\tau > T} \cdot p(\tau, y, z) d\tau \right) dz \\ &= \int_{g(D)} f(g^{-1}(z)) e^{h(z) - h(y)} \left(\int_T^{+\infty} p(\tau, y, z) d\tau \right) dz \end{aligned}$$

because $p(\tau, y, z) = 0$ for $z \notin g(D)$ and $\tau > T$: if $Y_T \notin g(D)$, then $\tau_Y \leq T$

$$= \int_{g(D)} f(g^{-1}(z)) e^{h(z) - h(y)} p^D(y, z) dz$$

where we define p^D as the restriction of p on $\{\tau > T\}$

At last we have

$$C = \int_{g(D)} f \circ g^{-1}(z) e^{h(z)-h(y)} p^D(y, z) dz$$

$$Z = f(g^{-1}(Y_T)) \mathbb{1}_{\tau_Y > T} \cdot e^{h(Y_T)-h(y)} e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_i}) \right\}$$

$$\mathbb{E}^{\mathbb{P}} [f(X_T) \mathbb{1}_{\tau_X > T}] = C + \mathbb{E}^{\mathbb{Q}} [Z \cdot \mathbb{1}_{N \geq 1}]$$

III Using q again

This step is crucial but quick thanks to our previous work in Chapter I, where we showed that, given the now ordered uniform points $U_{(i)}$, we had, in the barrier case $D =]L, +\infty[$:

$$\mathbb{Q} \left(\min_{[0, T]} Y \geq g(L) | Y_T \right) = \prod_{i=1}^{N+1} \left\{ 1 - q(TU_{(i-1)}, TU_{(i)}, Y_{TU_{(i-1)}}, Y_{TU_{(i)}}, g(L)) \right\}$$

where :

- $q(t, s, y_1, y_2, \lambda)$, defined in Chapter I, is the probability for the minimum of Y over $[t, s]$ to fall under the level λ , given the two endpoints $Y_t = y_1$ and $Y_s = y_2$
- $U_{(0)} := 0$ and $U_{(N+1)} := 1$

But now, we know by definition of τ_Y that $\mathbb{Q} \left(\min_{[0, T]} Y \geq g(L) | Y_T \right) = \mathbb{E}^{\mathbb{Q}} [\mathbb{1}_{\tau_Y > T} | Y_T]$ and we can define the new q^{BB} function by :

$$\begin{aligned} q^{\text{BB}}(\Delta, y_1, y_2) &:= 1 - q(t, s, y_1, y_2, g(L)) \\ &= 1 - \exp \left(-\frac{2}{\Delta} (g(L) - y_1)(g(L) - y_2) \right) \\ &\text{where } s - t = \Delta \end{aligned}$$

And we get the following :

$$\tilde{Z} := f(g^{-1}(Y_T))e^{h(Y_T)-h(y)}e^\eta \prod_{i=1}^N \left\{ \frac{T}{\eta} c(Y_{TU_{(i)}}) \right\}$$

$$\mathbb{E}^{\mathbb{P}} \left[f(X_T) \mathbb{1}_{\tau_X > T} \right] = C + \mathbb{E}^{\mathbb{Q}} \left[\tilde{Z} \cdot \mathbb{1}_{N \geq 1} \cdot \prod_{i=1}^N q^{\text{BB}} \left(TU_{(i)} - TU_{(i-1)}, Y_{TU_{(i-1)}}, Y_{TU_{(i)}} \right) \right]$$

IV Conclusion

The last formula gives us the price of the European barrier option without ever asking c to be bounded : the rejection experiment of Chapter I, which needed $0 \leq -c \leq B$, has been replaced by a Poisson-weighted expansion of $e^{\int_0^T c(Y_s) ds}$ [4], in which c only appears through a finite product of evaluations along the path. The decomposition into $C + \mathbb{E}^{\mathbb{Q}}[Z \cdot \mathbb{1}_{N \geq 1}]$ then isolates a first term, computable as an integral against a truncated density with the very tools of Chapter II, and a correction whose variance Yamada-sensei expects to be small compared to C .

What remains to be done is of two kinds. On the theoretical side, the variance of this estimator has to be studied : it is finite only under a condition relating η to the size of c , and the choice of η is a trade-off between the cost of a simulation – the number of Poisson points – and that variance. On the practical side, the formula has to be implemented and tested the way Chapter I was, against a scheme whose bias is known ; this is the work now in Naito-san's hands.

Conclusion

I What has been achieved

Over the ten weeks of this internship at Hitotsubashi University of Tokyo, the same question – how does one compute $\mathbb{E}^{\mathbb{P}}[f((X_t)_t)]$ without a time grid ? – was answered three times, in three different settings.

- **Chapter I** builds a complete, unbiased and **tested** Monte-Carlo estimator for a path-dependent payoff : Lamperti, Girsanov, removal of the stochastic integral, the Beskos-Roberts rejection trick, and finally the Brownian bridge together with the reflection principle for the barrier.
- **Chapter II** proves the decomposition (E) for the (CEV) model and carries the computation of $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}$ all the way to a closed formula, including the proof that C^{BS} is twice differentiable. It also shows why the correction term becomes small when β approaches 1, which is exactly the regime where the method is worth using.
- **Chapter III** removes the boundedness assumption on c through a Poisson expansion, and gives a formula for the barrier price in the general setting.

What I take away from these three chapters is less the formulae than the fascination I feel toward Yamada-sensei and researchers in general, who manage to find countless ideas and explore them in order to push further the limits of what humanity can achieve. When you think about it, I never really went out of the rails Yamada-sensei had already put in front of me before I even landed in Japan. How could he predict we would reach such promising results ? He gave me papers with questions I had to answer, like he knew from the beginning something big was buried beneath a thin layer we had to scratch. And he was right.

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III What comes next

Three directions are open, in increasing order of ambition :

1. **Implement and test** the Chapter III estimator the way Chapter I was tested, and measure the variance reduction that **(E)** was supposed to bring : those numbers are still missing.
2. **Study the choice of η** , which governs both the cost and the variance of the Poisson estimator, and more generally understand how far unbiased methods can be pushed before their variance makes them more expensive than the biased schemes they replace.
3. **Move to higher dimension**, where the Lamperti transform no longer exists in general, and where grid-free unbiased estimators would be the most valuable – in particular as generators of clean training data for the machine-learning approximations of high-dimensional PDEs that motivate this line of research [19, 20].

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